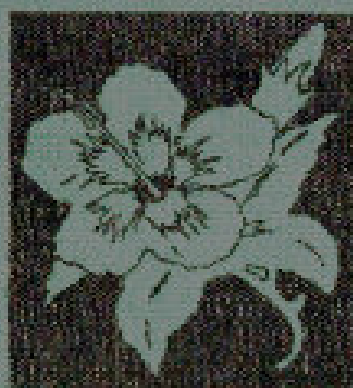


HAWAIIAN FLOWERS



LORAIN E. KUCK
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FLOWERS ***

HAWAIIAN FLOWERS

By
LORAIN E. KUCK
and
RICHARD C. TONGG

Illustrated by
TED MUNDORFF

TONGG PUBLISHING COMPANY, HONOLULU
1943

By the same authors
THE TROPICAL GARDEN
Its design, horticulture and plant materials.

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Preface



The working title of this book during its preparation was that question so often on the lips of tourists in Hawaii,

“What flower is that?”

Had there still been tourists in Honolulu, it is likely that that name would have been used on the cover, for the book is designed to help answer it. Now that the islands are filled with visitors intent on more serious business than pleasure, it has seemed best to call it simply “Hawaiian Flowers.” It is published at this time in the hope that some of these visitors may find in it an hour’s escape from the strain and pressure of war. For, in spite of war, the flowers still bloom in Honolulu, often right over the bomb shelters that fill gardens and parks.

In using the book to help identify the island flowers, it is hoped that the color plates and descriptions together, will make it fairly easy for the layman. Persons interested in more serious study will know how to go to more technical works for information. Care has been taken to have each name accurate to aid those going to the accepted authorities.

In making the color plates the attempt has been made to have them both artistic and scientific, a difficult undertaking under any circumstances. In striving to present pictures of individual flowers so they may be identified, it has been necessary, at times, to sacrifice the gorgeous effects presented in nature by masses of these flowers; such, for instance, as is seen on the Shower trees. The writers wish to take this opportunity to thank Ted Mundorff, the artist, for his trouble in making the reproductions as accurate as possible. ii

They wish, also, to express appreciation to Mr. Edward L. Caum, botanist with the Hawaiian Sugar Planters' Experiment Station, for reading the manuscript and checking the botanical information it contains. Mrs. Arthur Silverman, in the Library of Hawaii, has been most helpful in looking up obscure references.

Works consulted in preparing the book include Rock's "Ornamental Trees of Hawaii," and his "Indigenous Trees of Hawaii"; Marie Neal's book, "In Honolulu Gardens" and Bailey's Cyclopedia of Horticulture.

Honolulu, 1943

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Introduction

WHERE TO SEE HAWAIIAN FLOWERS?



If you are already in Hawaii, this question will seem purely rhetorical, since the answer is so obviously, “Anywhere.” There are few places indeed, in the islands, where flowers of some sort are not in sight. Mostly, of course, these are the Hibiscus, but you will soon notice others and the quest for new ones is apt to develop into a full-sized hobby. For the benefit of newcomers, a few of the gardens, parks and drives where you will find many Hawaiian trees and flowers are given here. By the time you have covered these places, you will know how to go on by yourself.

ROYAL HAWAIIAN HOTEL GARDEN

Few finer, tropical gardens can be found in the Islands than the one which forms a setting for the Royal Hawaiian Hotel at Waikiki. Not only does it contain a very large assortment of trees, shrubs and special tropical plants, many of them rare, but it has been laid out to make the very most of the

charm inherent in the words “tropical garden.” There are cool, green, jungle depths, gorgeous blossoms and wide shady lawns, all beautifully maintained.

To see these gardens it is best to start at the hotel steps, facing the coconut grove. This particular group of palms probably contains the oldest ones in Honolulu, some of the tallest, slender trees being estimated at well over a hundred years in age. Hawaiian kings kept their surf boards in the shade of these trees. In the early days, indeed, they were almost the only trees of any kind in the city. Old pictures show a dusty, sun-baked expanse between Honolulu harbor and Waikiki, with this grove standing up conspicuously. The luxuriant green growth of gardens which now mantles Honolulu is something comparatively recent in its existence, the result of a water system and an interest in gardens. 12

Near the steps and terrace of the hotel you will find plants with colored foliage including Crotons, red-leaved Tis and the Beefsteak plant. The fountain is surrounded with Lauae fern (*Polypodium phymatodes*) and near it grow the heart-shaped leaves of Caladiums and A’pes, (pronounced ah-pays). This is the Hawaiian word for various plants of the Taro family.

To the left of the fountain stands a magnificent specimen of a Monkeypod tree, throwing out its giant branches over a wide expanse of lawn. Beyond it is a Bengal banyan, (*Ficus bengalensis*), with bunches of aerial roots dangling from the branches. When they touch the ground they develop into secondary trunks.

The arbor is covered with Crimson Lake Bougainvillea. In the plantings around it you can find the shrubs illustrated on [Plate VI](#), for the flowers pictured on it were gathered at this spot.

Further along, to the right, is a Pink and White Shower tree, a Golden Shower and an Octopus tree. The latter, (*Brassaia actinophylla*), can be identified by its radiating, octopus-like arms, covered with dark red buds or blossoms. There is also a Hala or Pandanus in this area.

Walking across the lawn and turning left, you pass the thick border of Oleander and Hibiscus shrubs which screen out the street. In this section may be found a Royal Poinciana, a Potato tree, a Kukui and some Breadfruit

trees, (*Artocarpus incisa*). You can tell the latter by their very large, deeply lobed, oblong leaves, and the rough, round, green fruit, about the size of a muskmelon, which grows on the ends of the branches. This fruit, when baked, tastes something like a sweet potato. Here is also a Pomegranate tree and beyond are Bananas.

Turning again to the left you enter the real tropical garden, a cool, 13 green, jungle under the second coconut grove. Some of the trunks of these palms are enwrapped in the great green and gold leaves of the Pothos or Philodendron vine. Growing under the tall palms are lesser ones, along with bamboos and tree ferns. The latter have soft, fibrous, brown trunks with three or four very large, typical, fern-leaves growing out of the top. The paths hereabouts are bordered with more of the Lauae fern and with Begonias and Anthuriums, while green and red Ti plants and A'pes fill in the background. If you keep somewhat to the left, you approach the hotel building again, where you will find the Torch Gingers and Red Gingers serving as a base planting.

As you pass beyond the building you can see a Tiger's Claw tree and over to the right, near the road, a Plumeria. Planted around here are Spider Lilies and Shell Gingers. Near the side door of the hotel are Tecomarias or Cape Honeysuckle, and nearby is a glowing African Tulip tree.

The wide grassy cove that fills the curve of the building beyond is bordered by a very fine collection of Hibiscus. Trees in this area include the Yellow Poinciana, more Shower trees and, across the lawn, an old specimen of the Papaya. This has a smooth trunk marked by the leaf scars. The leaves grow out stiffly, on long stems, near the top. The fruit grows directly out of the main trunk.

If you cross the driveway and follow the walk, starting near the entrance, that leads to the tennis courts, you will pass a different assortment of trees and shrubs, those that like sun and not too much water. Date palms have thick, grey-green fronds and rough, stubby trunks. There are pink Plumerias, a Chinese Rice Flower tree, a Milo and Be-still trees. Near the tennis court is a Jacaranda and some Lime trees, and at the very end are both purple and white flowering Orchid trees. Near the Sports Office you can find Jatropha shrubs, the Caricature plant and the red and yellow Pride of Barbadoes.

Returning to the hotel you pass one of the most common trees in the island, the Kiawe or Algaroba, (*Prosopis chilensis*). It is a spreading, light-foliaged tree, often twisted picturesquely. It has fine, bipinnate leaves, small spikes of tiny, yellow flowers and yellow, bean-like pods. These trees now cover large areas of the island, although the seed of the original tree was brought here in 1828 by the Catholic missionary, Father Bachelot. They are natives of tropical America. 14

Bordering the drive at the rear of the hotel you can find the Lobster Claw, (*Heliconia*), and a specimen of the Kou tree. Growing on the large old Monkeypod trees that shade the motor turn-around are examples of the curious Staghorn fern.

DRIVING FROM WAIKIKI TO TOWN

The ride from Waikiki to the center of Honolulu provides an opportunity to see many more trees and flowers. If you go by the Ala Moana, you pass by Moana Park bordering the shore. It holds a building, with a large inclosed court, where the Flower Shows usually take place, and where the Park Board has its office.

If you go by way of Kalakaua avenue and King street, you will be interested in knowing that the line of trees growing in the parkway down the middle of Kalakaua for some blocks before reaching King, are Mahogany trees. At the corner of King and Keeaumoku is a park-like square with some fine, very old trees, including some of the Kapok, (*Ceiba pentandra*) which produces the kapok floss of commerce. This has huge surface roots, almost like flying buttresses. The building in this square is the office of the Territorial Board of Agriculture and Forestry.

A few blocks farther down King street is Thomas Square. In peace times you can watch the fountain in the center of this park as you sit under the shade of the four giant Banyans. There are specimens of the false Wili-wili here, which scatter their red seeds on the ground in early spring. Facing Thomas Square on the makai side is the private estate of the Wards known as the Old

Plantation, closed to the public. On the other is the Honolulu Academy of Arts, which is open to the public. 15

The Academy has two delightful court gardens in the building, one an Oriental, Chinesque garden, the other a Persian courtyard. At the back are a series of small formal gardens. There are always unusual flower arrangements in the Academy building.

Near the center of Honolulu is Iolani Palace, former home of royalty and now the capitol of the Territory. In the park around this building are many fine old trees and some interesting shrubs.

A short distance from the palace, up Punchbowl street, stands the Queen's Hospital. Years ago some unusual trees were planted in its grounds, including the Bombax. There is a good example of the curious Bottle tree or Baobab, (*Adansonia digitata*), near the entrance by the Emergency hospital. You can recognize it by its trunk, which quickly tapers from about five feet in diameter to hardly more than one foot, as various limbs branch off.

If you go straight out King street, past Fort Shafter, you will come to Moanalua Gardens, a park-like area at the foot of the hill. This is a private estate belonging to the Damon family, but the public is invited to visit it. There is a pond with tropical water lilies and some extremely large trees. There are also greenhouses filled with Orchids and Anthuriums which visitors are sometimes invited to enter.

NUUANU AVENUE

A ride out Nuuanu avenue, will reveal a wealth of trees and flowers. On the left hand side, shortly before reaching School street, you come to Foster Park. This old estate is now a public park, presented by Mrs. Mary Foster, but originally it was the garden of Dr. William Hillebrand. He was an early botanist and lover of flowers, who imported many new things into the islands and planted them here. The size of some of his old trees is now tremendously impressive. Besides this, the city has collected here many

unusual plants. It also maintains a greenhouse in which visitors can see Orchids in bloom and many other unusual, exotic plants. 16

Farther out, Nuuanu avenue is bordered by the fine homes of some of Honolulu's well-to-do citizens. From the road you can catch many glimpses of charming gardens. Farther up Nuuanu valley you run into a wild, natural growth. Here you can find Koa trees, yellow Gingers and green Tis.

MANOA VALLEY

Another interesting ride is through Manoa Valley, where there are many more homes and gardens. This takes you past the campus of Punahou school which holds a large number of interesting trees. Along its lower wall is the famous hedge of Night Blooming Cereus. These flowers have their blossoming period in July, August, and September.

If you follow Manoa Road, you will come to Waioli Tea Room, run by the Salvation Army Girls' Home. In the garden there is a real Hawaiian grass house.

Returning down Oahu Avenue you come to a small square, called Kamanele Park, which holds an unusual Rainbow Shower tree. A turn to the left here will take you into the grounds of Mid-Pacific School; at the back is the nursery of the Mid-Pacific Horticultural Establishment to which visitors are welcome.

A short distance beyond Kamanele Park is the campus of the University of Hawaii. This holds many unusual plants and trees, including the famous Sausage tree (*Kigelia pinnata*). It grows in the lower corner, across from the fountain.

DIAMOND HEAD AND KAHALA

Still another interesting drive is to go toward Diamond Head from Waikiki, passing through Kapiolani Park, with its row of Ironwood trees (*Casuarina equisetifolia*), like mainland evergreens. The road around Diamond Head passes many more fine gardens, and farther on you come to the residential area known as Kahala.



Plate I

HIBISCUS—CHAPTER 1



Identification key

- (1) *Coral Hibiscus*
- (2) *Double Yellow Hybrid*
- (3) *Waterfall Hybrid Hibiscus*
- (4) *Turk's Cap*
- (5) *Rose of Sharon*
- (6) *Common Red Hibiscus*
- (7) *Hawaiian Native White*



Plate II

FLOWERING TREES—CHAPTER II



Identification key

- (1) *Gold Tree*
- (2) *Royal Poinciana*
- (3) *Potato Tree*
- (4) *Tiger's Claw*
- (5) *Plumeria*
- (6) *Be-still*
- (7) *African Tulip*



Plate III

FLOWERING TREES—CHAPTER II



Identification key

- (1) *Golden Shower Tree*
- (2) *Pink and White Shower*
- (3) *Orchid Tree*
- (4) *Coral Shower*
- (5) *Rainbow Shower*
- (6) *Yellow Poinciana*



Plate IV

FLOWERING TREES—CHAPTER II



Identification key

- (1) *Monkeypod Tree*
- (2) *Bottlebrush Tree*
- (3) *Jacaranda*
- (4) *Chinese Rice Flower*
- (5) *Wong-lan*
- (6) *Bombax*

Chapter I

HAWAII'S OWN FLOWER—THE HIBISCUS



What is that flower seen everywhere, you may ask, the one with five, stiff, papery petals, and a column rising in the center? Such blossoms are seen tumbling in cascades of pink from small trees, spangling hedges with red polka dots and decorating green shrubs in an artificial way, looking like something designed for an old-fashioned stage set.

The answer, of course, is Hibiscus, Hawaii's own flower, and the outstanding flower of all the South Seas. You will see these blossoms worn in the hair of Hawaiian women, or tucked over the ear of Samoan men when they dance the native *siva-siva*. You will find them strewn down the length of leaf-covered tables prepared for a native feast or laid out in colorful exhibition on counters of staid Island banking houses. You may run into them formed into huge fountaining bouquets, the flower heads fastened to long, artificial stems, or, at shows and entertainments, you may see them covering walls and other objects solidly.

All these uses and more are possible because the Hibiscus possesses the unique trait of not wilting for a day after it is picked. These flowers open at

dawn and live but a single day. Whether they be left on the shrub, picked and put in water, or laid out dry on a table top, they remain fresh and crisp until nightfall, when they suddenly close. Fresh buds open every day, so the Hibiscus is always in bloom. 18

Individual Hibiscus flowers follow a plan of five, with five petals, five stigmas, five lobes to the calyx and the like. The stigma is branched into five parts and is usually a bright crystalline red, like a bit of coral at the top of the central column. Stamens grow on the sides of this column, yellowing it with their pollen. This central column is like the one seen in hollyhocks, a flower which is a relative of the Hibiscus. Usually it is stiffly upright, but sometimes it sweeps outward in a graceful curve. When this is the case it is good indication that the flower is a hybrid and had in its ancestry the Coral Hibiscus, which has a very long pendulous column.

Hibiscus also form double flowers, the pink ones sometimes suggesting old fashioned cabbage roses or peonies. They are formed when the stamens are modified into petals. Both single and double flowers appear in all hues but blue. Color tones vary from clear white through palest pink and yellow to glowing scarlet, orange and gold, deepening to richer tones and dark crimson. In recent years hybrids have been produced which bring several hues into a single flower, one with a red center, for instance, may have yellow petals bordered in pink. Some of these polychromes are more interesting than attractive. The size and shape of the flowers vary also. Average diameter of a blossom may be five or six inches but some are dwarfs of an inch, while selection has produced some a foot across. These large ones seem even more artificial than most of the Hibiscus, almost requiring to be touched before their reality is established.

The Hibiscus shrub is rather undistinguished in appearance, growing sometimes into a tall tree twenty feet high, but usually nearer eight or ten. The leaves are opposite, roundly pointed and often slightly scalloped.

The Hibiscus is the floral emblem of the Territory of Hawaii. It was so decided in 1923 by a joint resolution of the Legislature which designated it as “a beautiful, indigenous blossom which grows luxuriously on all the islands, appearing to be most generally representative, no other flower having so great a variety of color or form, or such 19

continuous blooming.” Botanists point out that in making this statement the legislature conveniently overlooked the 150 or so species of Hibiscus which are not native of Hawaii, in favor of the six or seven which are. Hawaiians called the native blossoms “Kokio.”

There are two species of white Hibiscus native to Hawaii, the commonest being *Hibiscus arnottianus* Gray. It is found on Oahu and Molokai. During long periods of time this species grew in isolated areas in the islands until it developed varieties which appear to be quite different from each other. These are known by such names as the Tantalus White, the Waianae White, the Punaluu White and so on. Another native white Hibiscus is *H. Waimeae* called the Kauai White, (which may, however, be only a form of the other.) These two are the only ones of all the Hibiscus which are fragrant, having a faint, delicate scent. This characteristic has sometimes been transmitted to its descendents, so scented hybrids will occasionally be found. These native whites also remain open longer than the single day which most Hibiscus flowers live, *Hibiscus Waimeae* sometimes lasting even three days. Occasionally, therefore, a hybrid will also be found which stays fresh for several days. Both of these characteristics are sought by hybridists.

Other native species of Hibiscus are *H. kokio*, which occurs in red or coral colors on Hawaii, Oahu, Molokai and Maui. *Hibiscus kahili* grows on Kauai in several forms colored red or pink, although this may be a variation of *H. kokio*. *H. brackenridgei* is a yellow species, sometimes with a purplish center, growing on Kauai, Oahu, Molokai and Maui, while *H. youngianus* is pink or purplish.

One of the best places to see many varieties of Hibiscus is the garden of the Halekulani Hotel. Mrs. Clifford Kimball has made the flower her 20 hobby for years and achieved remarkable results.

COMMON RED HIBISCUS. CHINA ROSE

***Hibiscus rosa-sinensis* Linnaeus**

The species of Hibiscus which seems to have the greatest number of variants is, botanically, *Hibiscus rosa-sinensis*, sometimes called the China Rose. The

type flower is scarlet, usually with a deeper colored throat and about five inches in diameter. It is the one oftenest seen in hedges, since the shrub serves admirably for this purpose.

These plants seem to have innumerable variations in color and shape, the former running in hue from yellow to crimson and appearing in both single and double forms. In fact, it seems that most of the plants brought into Hawaii, which were not themselves hybrids, were forms of *rosa-sinensis*. Although, most of these imported forms have usually been classed as variants of *rosa-sinensis*, it is now believed that many of them were hybrids. The remarkable variation which occurs when these forms are crossed points to a mixed ancestry. ([Plate I](#))

CORAL HIBISCUS

***Hibiscus schizopetalus* Hooker**

A species which stands out as markedly different from the usual Hibiscus is *H. schizopetalus* which has been given the name in Hawaii of Coral Hibiscus. This is doubtless because its deeply cut, turned-back petals suggest branches of red coral. The flower stem is very slender, so the weight of the head causes it to fall over and hang down, bell-like. And since the central column of this flower is extremely long and slender it sways far beneath the flower, giving an effect of peculiar grace. This plant has leaves that are small and fine. It is tall and slender, almost vine-like, so that it has been used to make arbors. The Coral Hibiscus has been a parent of many island hybrids, imparting to them its grace, its lengthened column and its frilled petals. ([Plate I](#))

HAWAIIAN INTEREST IN HIBISCUS

While the Hawaiians had always loved their flower and the Common Red seems to have been brought in at an early date, probably direct from China, real interest in Hibiscus culture began about the turn of the century. Around

1902, Walter M. Giffard began crossing different strains and getting some of the spectacular results for which this plant is noted. Interest grew and as Island people traveled, they often sent or brought home new varieties. One of the persons who became interested in the plant was Gerrit Wilder, who seems to have held the first Hibiscus show. This was in 1914, and he was able then to exhibit some 400 different varieties. In the years following, interest was very widespread and many people made crosses, until there were literally thousands of different forms and colors.

HYBRIDS

Crossing is very easy. The pollen of one flower is dabbled on the pistil of another, after its own pollen has been removed. Precautions are taken to keep the bees from stepping into the experiment, by protecting the crossed flower with a bag. The seeds ripen in a month or so and when planted may be expected to blossom in about a year.

The outcome is a grab bag of mixed and unexpected characteristics. By selection and care, some hybrids have been produced that are amazing. New shades and tints come into being, and mixtures of colors. Doubles become larger and curlier. No one knows how many thousands of these hybrids have been produced in Hawaii. Some of them are fairly well established and widespread by propagation through cuttings; others never existed in more than one plant. The only color no one has ever produced is blue. It is the hope of all hybridists.

WATERFALL OR BUTTERFLY HIBISCUS

One of the most conspicuous of the hybrids because of its prolific blooming is that sometimes called the Butterfly or Waterfall Hibiscus from its graceful form. This is believed to be a cross between the Native White and Coral. The plant often grows into a small tree and is usually a mass of pink flowers. As a rule, it is the only one of all the Hibiscus plants that

carries enough color to be outstandingly conspicuous in the garden. On others, the flowers are scattered. ([Plate I](#))

YELLOW HIBISCUS

Among the most beautiful and sought after Hibiscus are those with yellow or orange blossoms. The range of tone is wide, the colors appearing from palest lemon to rich yellow, gold and brilliant orange. The yellow flowering plants are usually rather small and carry only a few blossoms at a time. ([Plate I](#))

ROSE OF SHARON. ALTHEA

Hibiscus syriacus, Linnaeus

A close relative of the true Hibiscus is the lavender or white Althea or Rose of Sharon. The central column is white and the center of the flower usually dark red. It is a native of Syria and the Holy Land. ([Plate I](#))

TURK'S CAP

Malvaviscus arboreus Cavanilles

Another close relative of the true Hibiscus is the little red Turk's Cap. This looks like a small Hibiscus flower which has not opened. It never does open widely, but sometimes the number of the half furled flowers is so great the shrub appears quite red from them. ([Plate I](#))

Other relatives of the Hibiscus are the Hau tree (*Hibiscus tiliaceus*) and the changeable mallow, (*Hibiscus mutabilis*) not to mention the Okra and Roselle among edible plants.

Chapter II

THE BLOSSOMING TREES



Hawaii's most impressive floral displays are the flowering trees which line the streets and fill the gardens. They rival, in the masses of their flowers, the fruit trees of the mainland, but they possess colorings of tropical brilliance which have no rivals outside of warm countries. Moreover, instead of blooming for a few brief days, or at most, weeks, these trees continue in bloom for months at a time. Some of them never cease. About the middle of June, there is a period when the display is at its best. This is when all the shower trees are out, the early ones still lingering, the later ones coming on, and when the red of the Poinciana, and the blue of the Jacaranda, add their hues to the galaxy of yellows and pinks. At this time, too, the Plumerias bring white and rose to the scene. Midwinter finds most of these gone, but the African Tulip flowers are then a fiery crown on many trees and some trees, like the canary colored Be-still, continue their flowering all the year round.

GOLD TREE. SUNSHINE TREE

***Tabebuia donnell-smithi* Rose**

One of the very special events in Honolulu's floral calendar is the blooming of the Gold or Sunshine tree which grows on School Street, near Nuuanu. The time of year at which this takes place is highly uncertain, being sometimes in midwinter, again late in the spring. Certain other specimens of the tree are just as erratic, and none seems to have any relation to 24 another, so that if the flowering period of one is missed, another may be found in bloom.

When not in bloom the Gold tree is rather ungainly in appearance, its smooth, slender, light grey trunk lifting the branches high in the air far beyond reach. But when the flowers suddenly appear on the bare branches, it seems like the touch of Midas, for they are shining masses of the purest gold, or like crystallized sunshine. The breathtaking color is intensified by the necessity of looking up at them against the vivid blue of a spring sky.

The tree is a member of the Bignonia family, and individual flowers are of typical Bignonia form, a slightly irregular tube with five lobes, irregularly margined. The leaves which appear after the flowers fall are, on older specimens, of compound form made up of opposite leaflets. On young trees the leaflets radiate from a common center.

The chief specimen of this tree grows in the Foster Gardens, now a public park, but originally the home of Dr. William Hillebrand, who must have planted it along with the many other novelties which he introduced. A second tree now grows across School street in a small nursery garden and a fine specimen is found in Moanalua gardens. Still another grows in the school grounds across from the Nuuanu YMCA.

It is a native of tropical America. ([Plate II](#))

ROYAL POINCIANA. FLAMBOYANT

***Poinciana regia* Bojer**

Of all Hawaii's flowering trees, the Royal Poinciana is easily the most stunning and conspicuous for sheer color and brightness. A solid mass of red, it is, nevertheless, not merely gaudy, but one of the most graceful and picturesque of trees, a flat umbrella of color in small specimens, or composing into long sweeping curves in larger ones. It suggests the massive regalia of some magnificent Oriental potentate. The color is most dramatic if viewed against a grey, valley raincloud in the late afternoon, touched 25 by the level yellow rays of the declining sun.

The tree may become forty feet high, if growing under favorable conditions, but if the roots are cramped it remains quaintly dwarfed. Although bare for a short season in winter, the general flowering season of the Poinciana is long, for some trees begin to bloom early in spring, while others wait until late summer to open. June is the month when most of them are in bloom and the streets are consequently most gorgeous.

Individual Poinciana flowers, which have to be looked for closely to be distinguished in the masses of bloom at the end of the branches, have five petals. One of these is white, on the flag-red trees, or yellow on those tending toward scarlet coloring. These light touches give a piquant effect to the mass of color. Long, curved, brown pods hang on the tree for months after flowers and leaves have gone, and show that this tree is a member of the legume family.

The flowers usually appear on the bare tree before the new foliage comes out, but in a short time the leaves appear and for some weeks the green and red colorings remain together. The leaves are fernlike, bipinnate in form, with very small leaflets. Even when the flowers have fallen to form a carpet of red underneath the green tree, it is graceful and attractive.

It is a native of the island of Madagascar and was named for de Poinci, a governor of the Antilles in the 17th century. The French have given it another name, Flamboyant, by which it is widely known all over the tropical world and one which is particularly appropriate.

No visitor to Hawaii can miss the Poinciana, but the row that grows along Wilder avenue is perhaps the most effective, although individual large trees elsewhere may be more impressive for size and form. ([Plate II](#))

POTATO TREE

Solanum macrophyllum Dunal

The nightshade family, to which the potato belongs, is one of the most widespread of botanical groups. It contains however, but few trees. One of these is usually called the Potato tree, because its flower is similar in form to that of the common potato. It grows in Hawaii, but is not common although it grows in some gardens and in the Mid-Pacific Horticultural Establishment. While similar in form to the ordinary potato blossom, the flower of the Potato tree is comparatively large, being about two inches across. It has five joined segments which give it almost a pentagonal outline. The color is a rich purple-blue when it first opens, but this fades to a pale blue and then almost white. In the center is a golden yellow column made of the thick anthers.

The tree grows very rapidly. Its leaves are large, about a foot long, and irregularly lobed. They carry a few sharp thorns along the back of the midrib. The plant is a native of Mexico. ([Plate II](#))

TIGER'S CLAW. CORAL TREE. INDIAN WILI-WILI

Erythrina indica Lamarck

Tall trees, bursting into pointed red blossoms in midwinter and early spring, are appropriately called Tiger's claw or Coral trees. The flowers are a deep, rich, red, very striking on the bare trees at this season. They grow in long clusters which radiate horizontally on woody stems from the ends of the branches. Individual flowers break out of the split side of a pointed calyx. Fundamentally of the pea-type, these flowers have one petal much larger than the others, the general effect being that of a pointed claw or feline toenail.

The leaves, which appear soon after the flowers, are made up of three triangular leaflets. The pod is black and contains dark red seeds. The

branches are thorny.

This tree is a native of tropical Asia and a member of the legume family. There is a fine specimen growing in the grounds of Iolani Palace which 27 blooms in January or February. Others grow on Punchbowl street near Beretania, and in the grounds of the Central Intermediate School.

A closely related tree, which is native to the Hawaiian islands, is called the Wili-wili by the native people. Botanically it is *Erythrina monosperma*. It grows in dry places on the islands where it is conspicuous for its pale red, orange or yellowish flowers, similar in form to the Indian Tiger's claw. The bright red seeds of this Wili-wili were made into leis by the Hawaiians, but nowadays most of the red seed-leis are made from the *Adenantha pavonina*, known as false Wili-wili. Adenantha trees, which have unusual, curling pods filled with the bright red seeds, grow in Thomas Square where the seeds may be picked up in the Spring.

The wood of the Hawaiian Wili-wili is very light, and so was used for making the outrigger log of canoes. ([Plate II](#))

PLUMERIA. GRAVEYARD FLOWER. FRANGIPANI

***Plumeria acutifolia* Poiret**

One of the most popular of Hawaiian flower leis is made up of the thick, waxy flowers of the Plumeria. They are particularly successful for this purpose because they remain fresh for a long time and have a fine fragrance. The common name in Hawaii, Plumeria, or, as the lei-women say, Pumeli, is derived from that of Plumier the French botanist, but a mistake was made in spelling it so the genus is now properly designated as *Plumeria*, not *Plumiera*. Although this tree is a native of tropical America, it grows in India and in the temple gardens of Ceylon, where it is known by the romantic name of Frangipani. In Honolulu it is extensively planted in cemeteries, from which it is known as the Graveyard flower.

The flower varies from almost white to yellow, with yellow centers. Another species with rich, cerise, colored flowers is the *Plumeria rubra*. These two

have crossed and produced a wide range of flowers with pink or peach colorings. One hybrid, of particularly striking apricot color was produced by Gerrit Wilder. A specimen may be seen on Manoa Road near the top of Punahou hill. 28

The Plumeria tree has a few stiff, blunt branches. At their tips, the clusters of flowers appear in early spring, when the tree is still bare, creating a highly picturesque effect. A little later the foliage appears and the tree continues to bloom the rest of the year. The flowers have five, waxy petals joining in a short tube. The leaves, which also cluster toward the tips of the branches, are very long and pointed at each end, hence the name *acutifolia*. When cut, the stems exude a milky juice.

This tree is a member of the Periwinkle family.

BE-STILL. YELLOW OLEANDER *Thevetia nereifolia* Jussieu

A small tree, holding a scattering of trumpet-shaped, yellow flowers and marked by shimmering, narrow, light green leaves, is popularly called the Be-still tree. There seems to be no reason for this name, unless it is that the slender leaves are never still and the name is a sort of invocation. The flowers are a clear, satiny, yellow, with a delightful fragrance. They grow here and there all over the tree, and at all times of the year. A less common variety has pale, orange-colored flowers. The fruit is a nut which is poisonous. Green at first it turns brown, then black.

The tree never becomes very large, twenty feet being about its maximum height. It always has a fresh look, even when growing in dry places, due to the light color of the leaves. The slender foliage suggests that of the Oleander (*Nerium*), accounting for its specific name of *nereifolia*, and this association also accounts for the common name of Yellow Oleander, which, however, is entirely wrong.

The tree is a member of the Periwinkle family and a native of tropical America. ([Plate II](#))

AFRICAN TULIP TREE

Spathodea campanulata Beauvois

Large, fiery red flowers, like cups of molten metal, crown the high branches of the African tulip tree. This tree differs from many of the flowering trees in Hawaii by producing its flowers all the year round. There is a season in midwinter when they seem to be brightest and most numerous, but this may be due merely to lack of competition.

Individual flowers suggest a lopsided cup, with five irregular, frilled lobes. The edges of the corolla are a vivid yellow, and the inside of the cup is yellowish also, with red streaks. The flowers grow in circular masses of closely crowded buds, a few developing at a time, so that the tree seems to be ever-blooming. The flowers grow out of a spathe-like calyx from which is derived the generic name of *Spathodea*. They are followed by boat-shaped pods, some two feet long, which split open and spill out masses of shining, flaky, winged seeds.

The leaves are large and compound in structure, made up of three or four pairs, and an end leaflet. The leaves are dark green in color, leathery and with conspicuous veining.

The tree is a member of the Bignonia family and a native of tropical Africa. Specimens grow in Kamanele Park, Manoa. ([Plate II](#))

GOLDEN SHOWER TREE

Cassia fistula Linnaeus

Immense pendant clusters of large, yellow blossoms, hanging in grapelike bunches among the leaves explain the popular name of the Golden Shower tree. Although the foliage does not fall, the yellow blooms sometimes cover the tree so completely they overwhelm the leaves and make it look as if this

tree were the only thing in the landscape which is standing in sunshine, all else being shadowed.

Leaves are very large and compound in structure, each leaflet being two to six inches long.

The bright, golden, yellow flowers have five petals, clearly veined. 30 Like all the shower trees, which belong to the Pea family, the yellow flowers are built on the general plan of a pea blossom, but the five petals are very nearly of the same size and shape. From the center of the flower project the long curving pistil and some stamens. This pistil develops into a straight, cylindrical, black pod, sometimes three feet in length. It has given the name of Pudding-pipe to the tree in India. This long “pipe” is the Cassia pod of commerce, a cathartic being made out of its sticky brown pulp. The tree is a native of tropical Asia.

Golden Shower trees line both sides of Pensacola street between Lunalilo and Wilder avenues. They are at their best in June and July. ([Plate III](#))

PINK AND WHITE SHOWER TREE

Cassia javanica Linnaeus

(*Cassia nodosa* Hamilton)^{[\[1\]](#)}

Great feathery masses of unevenly tinted, pink flowers cover this small tree, suggesting in their luxuriance and variable coloring the apple blossoms of the temperate zone. The flowers grow on short branchlets, in what seem to be tufts of reddish stems, the tufts growing out of the main branches so close together these branches are completely enwrapped. The splendid effect of such inflorescence makes the Pink and White Shower tree one of the most important in Hawaii’s annual procession of blossoms. The tree is deciduous and flowers often precede the leaves, but these shortly appear, adding contrast to the total effect, with their fresh green. The leaves are of feather form with many pairs of rounded, medium sized, leaflets.

Each flower is made up of five petals, from the center of which grows a tuft of stamens. The calyx and stem is dark red. Each petal is palest pink or white, with deeper pink veinings, giving the name of Pink and White shower. Eventually, too, the pale pink fades adding to the variegated effect. The tree remains in bloom for months, with June as its peak. It never 31 becomes very large and is often quaintly irregular in form. Long, cylindrical, brown, seed pods hang on when flowers and leaves have fallen. It is a native of tropical Asia.

This tree is grown widely as a street tree in Honolulu, Piikoi street between Wilder and Lunalilo being a good place to see it. There are, also, some fine specimens along Nuuanu avenue. ([Plate III](#))

ORCHID TREE

***Bauhinia variegata* Linnaeus**

Exquisite lavender or white orchids, as beautiful as Cattleyas, seem to grow on the small Orchid tree. It is not, however, related to the real orchids, but is a member of the legume family. When covered with pure white flowers the tree is a splendid sight, but the lavender variety, which blooms more sparsely, shows more beauty when the individual flowers are examined. They resemble strikingly the real orchid, with a large main petal marked with purple, and four crepy side-petals. A bunch of white stamens grows from the center. The flowers appear in the cooler months, with spring as the finest season. They are scentless.

The leaves of this Bauhinia, as of others, are peculiarly shaped being deeply cleft into two rounded lobes, so that they suggest the wings of a green moth. Insects find the leaves very succulent, so they are usually full of holes, or completely eaten.

The tree is a native of India where its bark is used for tanning and dyeing, while its leaves and flower buds are used as a vegetable. ([Plate III](#))

A pink-flowering member of the Bauhinia family is called the St. Thomas tree and is *Bauhinia monandra*. This has similar lobed leaves and pink

flowers, both being smaller than on the Orchid tree. The main pink petal is dotted with crimson and the tree is very gay when in bloom. After the flowers fall it hangs full of pods.

CORAL SHOWER TREE. PINK SHOWER

Cassia grandis Linnaeus

Earliest of the shower trees to bloom is the one which has come to be known in Honolulu as the Coral Shower, or sometimes as the Pink Shower. (But not to be confused with the later-blooming Pink and White Shower.) The Coral Shower flowers during March, April and May, its soft rose color and general appearance somehow suggesting pink coral. In effect this tree is strikingly like the blossoming cherries with loose upright limbs covered with pink flowers.

The flower buds are particularly attractive being rounded, velvety balls of delicate, pinkish lavender. The flowers hang in short racemes from the branches completely covering them in good specimens. Like the pink and white they have five petals, but are smaller and more evenly colored. Stamens and pistils project from the center. The leaves follow the first blossoms closely, the new foliage being pinkish. Leaves are pinnate, with the leaflets rather large. The pods are cylindrical and dark brown.

Unlike the other showers, which come from Asia, this tree is a native of tropical America.

Liholiho street, between Wilder and Lunalilo is bordered with these trees and a fine specimen stands on Punahou campus. ([Plate III](#))

RAINBOW SHOWER

Cassia hybrida
(*Cassia javanica* × *Cassia fistula*)

It probably was inevitable that sooner or later someone should try to cross the Golden Shower with one or the other of the two pinks. Fortunately this first took place some years ago, so that today the many “Rainbow Showers” resulting from this cross may be seen in all their breathtaking loveliness. They are among the most beautiful of all the flowering island trees, and no two are alike, unless propagated by grafting. In general, the 33 inflorescence of these hybrids seems more numerous than on either of the parents, a result, no doubt, of combining the numerous flowers of the Pink and White with the spreading growth of the Golden Shower. At the height of their bloom, some of these trees appear to be almost solid with great fluffy masses of color.

Hues vary from palest cream and lemon yellow through all manner of peach and apricot tints to some that are a rosy orange. Flowers of individual trees often hold two tones, resulting in this variety of coloring. This is an effect that is enhanced frequently by the difference between the inside and outside of the petals, the buds being of a different color from the full blown flowers. There is a great difference, also, in the form of individual flowers.

There is considerable variation in the blooming period, but on the whole the Rainbow Shower trees come out later than the others, with July and August as the months of greatest bloom. They may be seen on Farrington street between Wilder and Beretania, and there are some fine individual specimens, one on Lunalilo street between Pensacola and Kapiolani, another on Makiki above Nehoa. A fine, clear, yellow flowering tree, sometimes mistaken for a pure Golden Shower, stands in Kamanele Park. ([Plate III](#))

YELLOW POINCIANA

Peltophorum inerme Roxburgh

A tree with many upright spikes of small, deep-yellow flowers, bright against the greenery of the fine-cut leaves is called the Yellow Poinciana, because it was once classed as a Poinciana. At the same time it is in bloom, it hangs full of reddish-brown pods which are one of its characteristic features. The flowers appear in the late summer and autumn, although there may be a second blooming period at some other time of year.

The flower buds are round and covered with brown, velvety down. This same down also covers the young growth and the midrib of the leaves. 34 Individual flowers have five crepe petals of about the same size. The larger, triangular heads of bloom, are made up of smaller clusters. The reddish brown pods which follow the flowers remain on the tree for a long time. They are thin and flat and hold three or four seeds. The leaves are bipinnate, composed of many small rounded leaflets. There is no period when the tree is bare.

This tree, a member of the legume family, is a native of Malaya and the East Indies. It grows widely in Honolulu, with several fine specimens in the grounds of Iolani Palace. ([Plate III](#))

MONKEYPOD TREE

Samanea saman (Bentham) Merrill

Those huge, wide-spreading trees—the largest trees in Honolulu—which in spring and summer are often covered with a thin film of pink flowers, are Monkeypod trees. Stately and massive, with rough, dark bark, the branches of these trees support a rounded canopy of leaves. It is a single layer thick and casts a light shade over an immense area of ground. There is a giant specimen in Moanalua Gardens; others grow in front of the Library of Hawaii, and traffic passes around another great tree in the middle of Vineyard street, near Nuuanu.

The flowers are like short tassels made up of tufts of silky, pink stamens. They grow on short stems in bunches near the ends of the branches, and cover the tree lightly during the spring and summer. They are followed by the thick, dark, pods, which hang on the tree until the following spring. Leaves are compound in structure, made up of opposite pairs of pointed leaflets which fold together in the late afternoon. The leaves fall in spring, and, together with the drift of falling flowers, which comes a little later, and the hail of old seed pods, give the Monkeypod the name of being the dirtiest of Hawaii's trees. Owners who never finish sweeping up under them always remark, however, that the beauty of the giant tree is worth the trouble.

This tree is a legume, a native of Central America and the West Indies, where its native name is *zaman*, from which its scientific name is derived. ([Plate IV](#))

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BOTTLEBRUSH TREE

Callistemon lanceolatus De Candolle

Long, cylindrical spikes of red flowers, very like the round brushes used to clean test tubes or bottles, have given its common name to this tree. The effect is created by tufts of red stamens. In most varieties the flower spikes grow upright, but on some, as shown on [Plate IV](#), they hang in swaying pendants. Their color is a fine, pinkish red, which contrasts strikingly with the greyish green of the foliage. The latter is narrow, pointed and fine.

The tree belongs to the Myrtle family and is a native of Australia. It is not yet very common in Hawaii, but examples may be seen on the University of Hawaii campus near Dean Hall. ([Plate IV](#))

JACARANDA

Jacaranda ovalifolia R. Brown

Since blue is the rarest color in the flower world, a tree which is a mass of blue is something that will hardly be overlooked. Yet the rarity of the blue coloring in the Jacaranda is but little more important than the beauty of the tree as a whole. It becomes a large tree, with light grey bark, and is covered with foliage, each leaf of which is almost as attractive as a fern. These bipinnate leaves are symmetrical in form with many tiny leaflets. They usually fall in late winter and early spring, and the tree is bare for a short time.

The flowers, which appear in large, loose, clusters at the ends of the branches, are individually shaped like bells, with two lips, one with two lobes, the other with three. Their color is a soft, lavender blue. The

blossoming period is erratic, varying from midwinter to early autumn, but on the whole it is most conspicuous in spring. On individual trees, this blooming period is not very long, but different trees vary as to the 36 season when the flowers appear, so that one may usually be found in flower. The blossoms fall in masses, repeating their color on the ground like a reflection of the tree above.

The seed pods are of curious shape, round and rather flat. They have been worn as costume jewelry, when lacquered in gay colors and attached to ribbons.

Jacaranda belongs to the Bignonia family and is a native of Brazil, where it got its name. A fine specimen grows on Punahou campus. Others are along Nehoa street, near Makiki, and on Manoa road at Kamehameha avenue, and also on Makiki Heights Road. ([Plate IV](#))

CHINESE RICE FLOWER. MAI SUI LAN

Aglaia odorata Loureiro

The tiny, round, yellow blossoms of the Chinese Rice flower tree probably suggest rice to the Chinese, although each floweret is considerably smaller than a grain of rice. They occur in clusters of hundreds, near the ends of the branches, each tiny flower a minute yellow ball which looks like a bud, but never opens wide. The blooming period is spring and summer.

The tree is rather small, spreading, and very attractive, being covered closely with glossy leaves of compound form. It is a member of the China-berry family and a native of China, from which it was undoubtedly brought directly to Hawaii by some returning traveler. It is still found growing mostly in the gardens of Chinese residents. There is a good specimen in the Mid-Pacific Horticultural Establishment. ([Plate IV](#))

WONG LAN

Michelia champaca Linnaeus

The intense fragrance and heavy, ivory-colored, waxen quality of the petals, indicates the relationship of the Michelia to the Magnolia family. Brought to Hawaii by the Chinese, it is still a great favorite with them. Older 37 women wear a blossom in their hair, like a bit of carved ivory, and men may slip a few buds into their shirt pocket where the fragrance can be enjoyed.

The pointed buds, about two inches long, grow upright in leaf axils near the ends of the branches. Each is encased in a “nightcap” type of calyx, which slips off as the flower opens. The narrow waxen petals are numerous and grow around the greenish pistil. The flowers are rather inconspicuous but easily found by the scent, which is heavy, sweet and rather musky.

The tree grows upright and never attains great size. Its leaves are glossy, leathery, rather pointed and about eight inches in length. ([Plate IV](#))

BOMBAX

Bombax ellipticum Humboldt, Bonpland and Kunth

A Bombax tree growing in the Queen’s Hospital grounds presents such a striking appearance when blooming that it has become almost as well known as the Gold tree on School street. Unlike the latter, however, which blooms at erratic times, the Bombax can be depended on to put out its blossoms in March and April. They appear on the bare tree, a few at a time, suggesting a bunch of pink egret plumes.

The bud, growing upright on the bare branch, is like a stubby cigar, rising from the calyx which is like the cup of an acorn. The bud splits into five parts, which peel backward like a banana, and curl into a spiral. These are the petals, purplish brown outside, silky white within. The conspicuous part of the flower is the great pompon of pink stamens, an exploding rocket of color. The stamens are about five inches long.

Except when in bloom the tree is inconspicuous. Its foliage is made up of five radiating leaflets. The tree is a native of South America and a member of the Bombax family.

The species is said to produce pods in which the seeds are embedded in a cotton wool. It is, indeed, closely related to the Kapok tree which produces the kapok of commerce. (For an example of this tree, *Ceiba pentandra*, see the Foster gardens.) But in Honolulu the single specimen of Bombax has never produced seeds nor has it been propagated by other means, so that the tree in Queen's Hospital grounds has no rivals in interest although others are known to be growing in the Islands. ([Plate IV](#))

Chapter III

NATIVE TREES



Long before the first human being landed on Hawaii, these islands were covered with a thick growth of trees, shrubs and smaller plants. Land which first rose above the surface of the ocean as hot lava, or as coral, slowly deposited on undersea volcanic rocks, gradually became covered with plant life. It was brought ashore by ocean currents, by birds and by the wind. Among the first plants, doubtless, was the coconut. The huge nuts of this tree, covered with their tough, thick husks, can float on the ocean for months, and after finally washing ashore will take root and grow. The roots are tolerant of brackish water so that they can become established near salt water lagoons.

In spite of what we know of the way in which such isolated bits of land as Hawaii became covered with plants, it remains a constant wonder that so many different ones arrived here without the help of men. For Hawaii has a very rich native flora that has been of great interest to botanists. Many of the plants found growing here are common to all the South Sea islands. Others have so changed, through mutation and self-selection, that totally new species have been created, unique to Hawaii. Trees are among the finest of

these native plants, some of them attaining huge size. A few of those most often seen, are illustrated and described here.

HAU
***Hibiscus tiliaceus* Linnaeus**

One of the strangest of the native trees is the Hau (pronounced “how”). It is found on all the South Sea islands and is, indeed, cosmopolitan in the tropics everywhere. This tree is often grown on Hawaiian beaches to cast shade on the sand by training it over an arbor. It is normally a creeping or procumbent tree, spreading along the ground. Its long, sinuous branches interlock, if not trained, and eventually form jungles too thick to penetrate except by cutting. When this mass of branches is lifted off the ground, by being trained over some strong support, they form a thick green roof which keeps out light showers. These arbors are called *Hau lanais* in Hawaii. They have been made at many places along Waikiki beach, one of the best being on the seaward side of the Halekulani hotel. Here the central tree is very old, its trunk, gnarled and twisted and several feet through, creating an unusually impressive specimen.

The Hau is a true Hibiscus, its flowers having the typical form with a central column rising from the center of the five petals. When the petals unfold they are a bright yellow color, usually with a dark spot at the base. As they grow older during the day they turn to an apricot color and when they finally fall, they are a deep red. The leaves are heart-shaped, green and leathery above, whitish and silky with hairs, beneath.

The bent branches of the Hau furnish the crooked sticks used to attach the outrigger log to native canoes. And as might be expected of such a unique tree, many legends cluster around it. ([Plate V](#))

HALA. LAUHALA. PUHALA. SCREWPINE
***Pandanus odoratissimus* Linnaeus**

Another plant of strange and curious appearance is the Hala, or Screwpine. It might be taken at first sight for a palm, since its leaves have the tough, 41 fibrous quality of palm leaves. Actually it belongs to a family which takes its name from this genus, the Pandanaceae. It grows all over the South Seas and the East Indies, and India.

The descriptive name of Screwpine comes from the way in which the long, narrow, spiny-edged leaves grow out of the branches, in winding whorls. On young specimens they often create curious spirals. On older trees the leaves have a tendency to form tufts at the ends of the branches. These branches divide in pairs, the tree forming a series of ascending Y's.

The most striking feature of this tree is the way in which it sometimes puts out aerial roots to support the main trunk. These roots grow out and downward, as stiff and straight as a stilt, propping up the tree in what seems an entirely artificial manner. Such roots seem to appear only when the plant needs them, when it is old and heavy or when the soil is moist and loose and does not offer a firm hold.

Male and female flowers grow on different Hala trees. The male flowers, called *Hinano Hala*, appear as a long, white, pendulous cluster of blossoms, the showy portion being the bracts. They have many stamens and the flower is extremely fragrant. In Hawaii these male flowers are not so frequently seen as the female. The latter appear as a solid, round ball at the ends of the branches, looking a good deal like the fruit of a pineapple, (see [Plate V](#)). It is a standard joke to point these out to tourists as proof that pineapples grow on trees.

The Hala fruit is a drupe, the various sections colored orange and very smooth and shining. When separated, the sweet scented fleshy part is strung into leis by the Hawaiians, making one of the most curious of these native garlands.

The long, fibrous leaves are called Lauhala and are split and woven into many products. Lauhala floor mats are frequently seen, while finer work is done in making purses, hats and fans.

A relative of the Hala tree is a vine (*Freycinetia arnottii*), called *Ie-ie* in Hawaiian. Its male flowers are a bright scarlet, and form a conspicuous sight in the cooler mountain forests where it grows wild.

OHIA LEHUA

***Metrosideros polymorpha* Gaudichaud**

The favorite flower of old Hawaiian song and legend is the *Ohia Lehua*. It is not found commonly at the warm levels of Honolulu, and never becomes more than a shrub there, but it may be seen in upper Nuuanu valley and on Tantalus. This plant reaches its greatest perfection, as a magnificent tree, often a hundred feet high, at the cool level of the volcano on the Big Island of Hawaii. The Lehua blossom is the special flower of that island.

The flowers appear as bright red pompons of stamens; sometimes however, they are white, pink or yellow, for the plant is extremely varied. The leaves are small and often reddish when young.

The Hawaiian people believe that this beautiful red flower is sacred to Pele, the goddess of the volcano. If a flower is picked she may show her anger by sending rain, but if an offering of a flower is first made to her, then others may be taken safely. The flowers are made into leis—the Sweet Lei-Lehua of the song—which are highly regarded.

The Ohia wood is dark, hard and very beautiful, much used for floors and fine carving. The early Hawaiians made images of their gods from it, as well as spears and other implements.

The Lehua is a member of the Myrtle family and found on various South Sea islands. ([Plate V](#))

Another shrub, (*Calliandra haematoma*) has been introduced into Hawaii and is now known as the *Haole Lehua*, that is, the foreign Lehua. It has tufts of red flowers, very like those of the native plant, but they are larger and even more showy than the true Lehua. During the winter and spring,

leis are made of these and are often seen, being casually called Lehua leis.

KAMANI ***Calophyllum inophyllum* Linnaeus**

A tree that grows wild along the seashore is given the name of *Kamani* by the Hawaiians. It has large, thick, leathery leaves, very smooth and shining, and clusters of waxy, white flowers. These flowers have four white petals and in the center is a mass of golden stamens surrounding a red pistil. They are fragrant.

The flowers are followed by round, reddish fruits which contain an oily nut. In other parts of the tropics, especially Fiji and India, this oil has important uses, but it is not extracted in Hawaii. The tree belongs to the Mangosteen family and is found widely through the tropics. The nut may have floated to the shores of Hawaii or possibly some early Polynesian voyager brought it in his double canoe.

True Kamani trees grow on the campus of the University of Hawaii. They may also be seen on a ride around the island, growing near the beach. ([Plate V](#))

A tree called locally False Kamani, is the *Terminalia catappa*. Its large leaves and the fact that it, too, grows near the sea probably caused it to be given this name. The special characteristic of this tree is the way its large leaves turn red before they fall in winter. A scattering of these big red leaves on the tree is very noticeable, and when the whole tree finally turns red it becomes a conspicuous object. New, shining green leaves shortly follow the old. A fine example of this tree grows in the grounds of Iolani Palace.

KOA ***Acacia koa* Gray**

Like the Ohia, the Koa tree does not grow well at lower altitudes, but at proper elevations, such as near the volcano, on the Island of Hawaii, it becomes Hawaii's largest and finest tree. It is, perhaps best known to town dwellers by its wood, known as Hawaiian mahogany, which has been extensively used in furniture. In cooler sections of Honolulu, such as Tantalus, and upper Nuuanu, Koa trees may be found growing. They seldom attain much size, but are often of very picturesque form.

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The most characteristic thing about the Koa is its sickle-shaped leaf, like those found on many other members of the Acacia group. It is not a true leaf, however, but the flattened, grayish-green petiole, or leaf stem, modified to serve as a leaf. Such a modification is called a phyllode. The true leaves grow on young trees and sometimes appear on new growth. They are bipinnate, or fernlike, with many small leaflets. Both true and modified leaves are shown in [Plate V](#), but no flowers. Koa flowers are very inconspicuous, being small, creamy balls of stamens and pistils, like so many of the other Acacia flowers. They are followed by numerous small, brown pods.

The tall straight trunk of the Koa tree was used by the early Hawaiians to make canoes. A fine tree growing in the mountain was selected by the *kahuna*, or wise man of the village, and laboriously cut down with stone adzes. Then the whole village turned out to help drag it to the water's edge. Here it was hollowed and shaped with stone implements. Finally, it was colored and the out-rigger attached, the entire process taking months, perhaps years.

MILO

Thespesia populnea (L.) Correa

A tree which has heart-shaped leaves, like those of the Hau, and Hibiscus-shaped flowers, but is of upright, normal form, is the Milo. The flower is a paler yellow than the Hau blossom, and has a red spot at the base of the petals. As it fades, it turns from yellow to a purplish pink. It is followed by a five-parted green capsule, which turns dark brown and hangs on the tree a long time.

The Milo is a member of the Mallow family and closely related to the true Hibiscus. Like the Hau, it is at home over a wide area in the South Seas and the Asiatic tropics. It is found growing on the beaches and is said to have shaded the grass hut home at Waikiki of the first King Kamehameha. ([Plate V](#))

KOU *Cordia sebestena* Linnaeus

A native tree with bright orange colored blossoms is called *Kou* by the Hawaiians. Botanically it is *Cordia subcordata*. Though found on other South Sea islands, it is now rare in Hawaii, while a close relative, the *Cordia sebestena*, is generally called Kou. This foreign Kou, which has been introduced from tropical America, is quite similar in general appearance to the native species. Its flowers are a rich, orange-red, about an inch across. They are tubular, with six broad lobes, frilled and crepe-like in texture. Opening in clusters of three or four at a time, they make a gay showing over the small tree on which they grow. The leaves are very rough and shaped like a heart. Several trees grow on Young street between Piikoi and Pensacola streets. ([Plate V](#))

The wood of the native Kou was highly prized for making the wooden bowls from which the Hawaiians ate.

KUKUI. CANDLENUT TREE *Aleurites moluccana* (L.) Willdenow

Conspicuous for its light colored foliage, groves of the Kukui tree are easily seen on the mountain side, from a long distance. The trees grow best in sheltered ravines and gullies, so that from a distance, the shadow of such ravines is usually lightened by the blotches of greyish green which mean Kukui trees. The trees grow down to sea level, however, and may be found

at many places in Honolulu, among others in the Royal Hawaiian Hotel garden.

Leaves of the Kukui tree vary greatly. One type has pointed lobes, another has lobes which have flattened into almost regular form. The leaves are covered with a grayish down, which is particularly heavy on the under side, giving the tree its light appearance. 46

The white flowers are very small with five petals. They come in soft, massed clusters, male and female flowers growing separately on the same tree. The male appear on the upper branches, the female on the lower. They are not very different in appearance, but the male (illustrated on [Plate V](#)) has slightly broader petals. The Kukui is the special flower of Molokai.

The round green fruit contains a nut which was very useful to the early Hawaiians. Containing a high percentage of oil, it was used to make a candle-like torch, by stringing the nuts on the slender midrib of a coconut leaflet. Sixty or seventy nuts would burn an entire evening. The nut is also edible, although purgative in its action especially when not roasted. A pinch of ground, roasted Kukui nut is always served at luaus.

This tree is a member of the Euphorbia family. It grows widely over the South Seas.

Chapter IV

TROPICAL SHRUBS



Hawaii has a wealth of flowering shrubs, most of them introduced from other tropical regions to augment the few which are native. These shrubs make up the larger portion of island gardens, since many of the annuals and perennials of the temperate zone do not grow well in a warm climate. Many of the shrubs have unusual flower forms and others create spectacular effects with masses of color.

PAGODA FLOWER

Clerodendron squamatum Vahl

Brilliant scarlet flowers, in large, loose, upright heads, proclaim the Clerodendron. The blooming period is winter and spring. Individual flowers are slenderly tubular, widening into five narrow lobes which turn back against the tube. The stamens and pistil curve beyond the flower in a small

red tuft. Stems of the flower head are also red and hold this color even when the green berries turn blue-black.

The shrub grows about ten feet tall. Its large, heart-shaped leaves are thick and velvety, with wavy margins and prominent venation. The stems and pedicels are downy, and the latter have a tendency to turn red as the leaf matures.

This exotic looking shrub comes from South China and India. It is a member of the Verbenaceae family. Specimens grow in the Royal Hawaiian Hotel garden and Thomas Square. ([Plate VI](#))

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GOLDEN DEWDROP ***Duranta repens Linnaeus***

The popular name, Golden Dewdrop well describes the clusters of small, bright, yellow berries which hang on this shrub a large part of the year. They are so plentiful they usually cause the slender, grey-stemmed branches to droop gracefully. They lend themselves to interesting arrangements. The shrub may attain ten feet in height. Its small, light-green leaves are pointed at either end.

The flowers are a delicate, lavender-blue, or white, very small and formed as minute tubes, with five lobes. They grow as spreading clusters at the ends of the branches.

This shrub, a member of the Verbenaceae family, is a native of tropical America. It is grown in the Royal Hawaiian Hotel garden. ([Plate VI](#))

SHRIMP PLANT ***Beloperone guttata Brandegees*** ***(Beloperone nemorosa)***

Rosy or yellowish bracts, overlapping with scale-like precision to form a curving tube, are highly suggestive of the curved tail of a shrimp and explain the popular name of this plant. The true flowers appear, one or two at a time, from beneath the colored bracts, near the tip. They are small, white, tubular, with purplish dots on the larger of the two lobes.

The plant is herbaceous and sprawling, growing at most to five feet. Its leaves are slightly rough to the touch, opposite, and of medium size. It is a native of tropical America and a member of the Acanthus family. It grows in the garden of the Royal Hawaiian hotel. ([Plate VI](#))



Plate V
NATIVE TREES—CHAPTER III



Identification key

- (1) *Hau*
- (2) *Hala*
- (3) *Ohia Lehua*
- (4) *Kamani*
- (5) *Koa*
- (6) *Milo*
- (7) *Kou*
- (8) *Kukui*



Plate VI
TROPICAL SHRUBS—CHAPTER IV



Identification key

- (1) *Pagoda Flower*
- (2) *Golden Dewdrop*
- (3) *Shrimp Plant*
- (4) *Flowering Eranthemum*
- (5) *Madagascar Periwinkle*
- (6) *Rondeletia*
- (7) *Star Jasmine*
- (8) *Galphimia*
- (9) *Plumbago*
- (10) *Ixora*



Plate VII

TROPICAL SHRUBS—CHAPTER IV



- Identification key
- (1) *Candle Bush*
 - (2) *Coral Plant*
 - (3) *Lipstick Plant*
 - (4) *Brunfelsia*
 - (5) *Cotton*
 - (6) *Ilima*
 - (7) *Thunbergia*
 - (8) *Mock Orange*
 - (9) *Justicia*



Plate VIII
TROPICAL SHRUBS—CHAPTER IV



- Identification key
- (1) Dwarf Poinciana
 - (2) Chenille Plant
 - (3) Cape Honeysuckle
 - (4) Crepe Myrtle
 - (5) Crown Flower
 - (6) Beach Naupaka
 - (7) Purple Lantana
 - (8) Pikake
 - (9) Crown of Thorns
 - (10) Singapore Holly

FLOWERING ERANTHEMUM
Pseuderanthemum reticulatum Radlkofer

Conspicuous for its yellowish leaves and small white and purplish flowers is this *Eranthemum*. The yellow color appears extensively on the young leaves and survives on the older ones as yellow venations, making the plant appear very bright and sunny. The tubular flowers grow in small spikes, the tubes broadening into four lobes. These are spotted with purple dots where they begin to broaden.

The plant grows about six feet high. It belongs to the *Acanthus* family and possibly comes from Indo-Malaya or Polynesia. Specimens may be seen in

the Royal Hawaiian Hotel garden. ([Plate VI](#))

VINCA ROSEA. MADAGASCAR PERIWINKLE

Lochnera rosea (L.) Reichenbach

Rosy pink or pure white flowers, on a short, herbaceous plant which looks like a temperate zone annual, is the Periwinkle. It is a cousin of the blue myrtle, or periwinkle of the mainland, but resembles the latter only in the general form of the flowers. Known elsewhere as Madagascar periwinkle (although not a native of that island) it is called in Hawaii simply Periwinkle or Vinca.

The flowers are flat, five-petalled and ever-blooming. Some of the white ones have a cerise eye in the center. The leaves are greyish, long, narrow, with blunt tips. It is a cosmopolitan in the tropics and can be found growing on the University of Hawaii grounds. ([Plate VI](#))

RONDELETIA

Rondeletia odorata Jacquin

Small, round, heads of many bright, red and yellow flowers mark the Rondeletia. It is a shrub which is not common in Hawaii but may be found in the Royal Hawaiian and Foster gardens. The flowers are tubular, with a bright yellow throat and orange-red lobes. The leaves are opposite and sessile. The shrub grows to about six feet high. It is a native of Mexico and a member of the Coffee family. ([Plate VI](#))

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STAR JASMINE

***Jasminum pubescens Willde*
*(Jasminum multiflorum)***

The Star Jasmine takes its name from the starry, white flowers which cover the plant at all seasons. Each has from four to nine, pointed lobes, radiating from the mouth of the slender tube. Ordinarily, they are scentless, but a variety with faint fragrance is now becoming popular. When the flowers fall, they leave a group of coarsely hairy, green calyxes like small green pompons. The plant is at first a sprawling shrub, but later becomes a vine. Its leaves grow in opposite pairs along the length of the shoots. They are pointed and slightly velvety. This plant is widely used in Honolulu. It is a native of tropical Asia, and like all the Jasmines, belongs to the Olive family. ([Plate VI](#))

GALPHIMIA
***Thryallis glauca* Kuntze**
(*Galphimia glauca*)

Small, terminal, clusters of bright, yellow, little flowers with red stamens, characterize the Galphimia shrub. It grows about five feet high and has small, glossy, opposite leaves. The yellow flowers bloom most of the year, making the plant popular in gardens.

The word Galphimia, by which it is popularly known, is an anagram of Malpighia, the name of the Italian physician for which the family to which it belongs was named. It is a native of southern Mexico. ([Plate VI](#))

PLUMBAGO
***Plumbago capensis* Thunberg**

The pale blue flowers of the Plumbago are frequently seen especially in dry places where the plant thickly covers the ground or grows over walls and trellises, often with a heavy undergrowth of dead branches below. The flowers have a long tube and five lobes, and grow in short clusters. Their blue color varies from a soft azure to palest tints, and there is a pure white variety. The species with pink blossoms is *Plumbago rosea*.

The blue flowering plant is a native of South Africa, near the Cape of Good Hope, as its name indicates. It gives the name of Plumbago to its family. ([Plate VI](#))

IXORA

Ixora macrothyrsa Teijsmann and Binnendijk

Large, round “snowball” heads of scarlet bloom make the Ixora a very conspicuous shrub. Sometimes it seems almost like a small tree for it can reach fifteen feet in height. The small individual flowers have four petals growing at the end of a slender tube that appears to be a red stem. These flowers are sometimes laboriously strung by the Hawaiians into leis, which become solid red cylinders, two to three inches in diameter. There are also plants with pale red or with white flowers, the latter scented. The shrub has fine, large, glossy foliage suggestive of that of the Coffee, of which family it is a member. It is a native of Malaya.

There is a tendency in Honolulu to mispronounce the name as if it were spelled Exoria. ([Plate VI](#))

CANDLEBUSH. ACAPULCO

Cassia alata Linnaeus

Upright stalks of bright yellow flowers, almost cylindrical in form, explain the popular name of this shrub. When first opening, the individual flowers are closer together on the stalk than shown on [Plate VII](#), heightening the suggestion of golden candles growing over the shrub. The flowers are at their best in winter, although they may be found later. The individual flowers are pea shaped, as the plant is a member of this family.

The leaves are luxuriant, each one from eighteen inches to two feet long. The leaf is made up of many pairs of large leaflets, which

increase in size from the base to the tip. The plant is probably a native of tropical America, but is widely grown in the tropics.

CORAL PLANT

Russelia juncea Zuccarini

Tiny, tubular red flowers, growing loosely over the drooping sprays of a graceful bush have again suggested coral and given the name of Coral plant to this shrub. The sprays may grow to six feet, with stems and a few leaves of greyish green. Individual flowers are about an inch long, tubular, and hang loosely on slender angular stems. The plant is a native of Mexico and belongs to the Figwort family. ([Plate VII](#))

LIPSTICK PLANT

Bixa orellana Linnaeus

Seedpods of the lipstick plant are extremely ornamental and are often used as dried floral material. They are covered with heavy soft, dark hairs, deep red when fresh and turning to stiff brown as they dry. The pointed pod splits to reveal rows of seeds covered with a red powdery material. This red covering provides the annatto dye of commerce, used among other things for coloring oleomargarine, butter and cheese. It is not produced commercially in Hawaii, but the plants are grown for this purpose in tropical America, where they are native. The name Lipstick plant was produced locally on the spur of the moment when a name was needed, based on the way the red material smeared the skin.

Flowers of this plant, which appear in summer, are pale orchid pink, the five petals surrounding a central mass of lavender stamens. The foliage is dark green and prominently veined. The shrub may attain almost the proportions of a small tree. Specimens grow in Iolani Palace grounds across from the Library of Hawaii. The plant is a member of the Bixa family. ([Plate VII](#))

BRUNFELSIA

Brunfelsia hopeana Bentham

A shrub curiously covered in spring with both blue and white flowers, superficially suggesting pansies in form, is the Brunfelsia. The two colorings are due to the fact that the flowers are a soft lavender blue when they open but fade to almost pure white before they fall. They have five velvety petals, which are actually the lobes of a slender tube, and they give off a delicate fragrance.

The shrub is woody, with light grey bark and sparse dark green leaves. This species is a native of Brazil. A specimen grows on Metcalf street near Hunnewell. ([Plate VII](#))

Another Brunfelsia grows in Honolulu with white flowers which turn a deep cream as they grow older. This is *B. americana*, a native of tropical America. It is almost scentless by day, but develops fragrance after dark. Both are members of the nightshade family.

COTTON

Gossypium barbadense Linnaeus

The cotton plant is a cousin of the Hibiscus and in Hawaii grows to be a tall shrub which is often used in gardens. The yellow flowers are formed like Hibiscus, but do not open widely. As they fade, they become tinged with purple, a color change which suggests the related Hau and Milo blossoms. The seed case is large, round and pointed and partly covered by three fringed bracts. When it opens the boll of white cotton, in which are the seeds, breaks out. This fluff of white fiber remains on the plant for a long time, suggesting a different kind of blossom. In Hawaii the local cotton is sometimes gathered and used for homemade mattresses, but it is inferior in quality.

Cotton leaves are heart-shaped and velvety, due to whitish hairs which cover them and give a greyish green appearance to the plant. ([Plate VII](#))

ILIMA
Sida fallax Walpers

Another close relative of the Hibiscus is the Ilima, a native plant which has orange flowers about an inch across, looking like miniature Hibiscus blooms. The color ranges in tone from light yellow through orange and buff, to brownish red, but the orange color is the most popular. Leaves and stems are covered with whitish hairs creating a velvety effect. The plant grows as a small shrub, rather straggling in appearance. It is seldom cultivated except by the lei makers, but is found wild in dry places. It also grows on other South Sea islands.

Leis made of these thin, silky flowers are the originals of the often-seen orange paper leis, like paper ropes about an inch in diameter. The paper leis, which are so artificial looking, are nevertheless really very much like the flower originals. The true Ilima lei has the soft texture of flesh, created by great numbers of the flowers being strung flatly together. In early days such leis were reserved for royalty and they are still called the royal lei. The Ilima is the flower of the Island of Oahu. ([Plate VII](#))

THUNBERGIA
Thunbergia erecta T. Anderson

Rich purple-blue flowers with golden throats are Thunbergias, closely related to the white and blue flowering vines of the same name. The velvety purple petals are lobes expanding a tube that is whitish without and bright yellow within. The flowers grow singly in leaf axils, emerging from a pair of whitish bracts. The blossoms are thin and delicate and fade almost at once after being picked.

There is a pure white flowering form with yellow throat.

The plant is an open, rather straggling shrub, about five feet high. Its slender branches bear small opposite leaves, rather pointed at either end. It is a native of tropical west Africa and belongs to the Acanthus family. ([Plate VII](#))

MOCK ORANGE

Murraya exotica Linnaeus

The Mock Orange bears flowers several times a year in sporadic outbursts which cover the plant with clusters of small snowy blossoms. At such times the fragrance nearby is so intense that no one in the vicinity can miss it. The Mock Orange is a true member of the Citrus family and has the delightful scent which they all possess. Individual flowers are five petaled, and waxy, like a very small orange blossom. When these petals fall they cover the ground with white. The fruit is a small red ball filled with a large seed. It is not edible, but very bitter.

Between periods of bloom the shrub, which may become a small tree, remains attractive in its small, glossy, dark green foliage. Leaves are compound, the leaflets being about two inches long and pointed. This shrub is a native of tropical Asia and widely grown. ([Plate VII](#))

JUSTICIA

Odontonema strictum Kuntze ***(Justicia carnea)***

The Justicia bears stiff, upright spikes of red flowers, each one a waxy little cornucopia with five small lobes. The flower stalk is a mass of buds, of which only a few, up and down its length, develop at a time giving it a rather ragged and irregular appearance, but prolonging its blooming season almost indefinitely. The open flowers fall quickly.

The plant is herbaceous and grows about five feet high. Its leaves are bright green and glossy, large, pointed, and with prominent veins. It is a native of tropical America and belongs to the Acanthus family. ([Plate VII](#))

DWARF POINCIANA. PRIDE OF BARBADOES
***Caesalpinia pulcherrima* (L.) Swartz**

Bright clusters of fiery scarlet and yellow flowers growing on the higher branch tips of a tall shrub or small tree, announce the Dwarf Poinciana. 56 While not a true Poinciana, it is a close relative, so that its common name is not far amiss. Individual flowers are smaller, but quite similar in form to those of Royal Poinciana, with five crepy spreading petals and a colored calyx. The petals are sometimes margined with yellow which gives added brilliance to the effect. Very long stamens and pistil project from the center. Unlike Poinciana, the plant blooms most of the year. Flat seedpods follow the flowers. There is an all-yellow flowered form.

The leaves are doubly compound, with many small rounded leaflets along the pinnae. It is a legume, and widely grown in the tropics. In India it is the sacred flower of Siva. Specimens grow in the University of Hawaii grounds. ([Plate VIII](#))

A similar plant with flowers of bright yellow but with conspicuous, long, red stamens is *Caesalpinia gilliesii*, sometimes called Bird of Paradise.

CHENILLE PLANT
***Acalypha hispida* Burmann**

One of the strangest looking of tropical shrubs bears long, thin velvety tails of dark red, which well deserve the common name of Chenille plant, or sometimes, Redhot Cat-tails. The shrub will attain eight feet and presents a striking appearance with these long, crimson, flower spikes hanging from among the large green leaves. The tails are made up of the staminate flowers,

which have no petals; pistillate flowers are inconspicuous. A fine plant grows in front of Gumps at Waikiki. The shrub is a member of the Euphorbia family and a native of the East Indies. ([Plate VIII](#))

CAPE HONEYSUCKLE
Tecomaria capensis (Lindley) Seemann

The orange red flowers of this shrub have the typical trumpet form of the Bignonias, to which family it belongs. The slightly curving tube 57 broadens into five lobes, beyond which extend the yellow stamens. The flowers appear in small clusters at the branch ends.

The shrub is sprawling and vinelike, often used as a ground cover. It may be seen at the University of Hawaii. It has small, compound leaves notably dark green in color, each leaflet having a serrated margin. Its specific name, *capensis*, reveals its native home to be South Africa.

CREPE MYRTLE
Lagerstroemia indica Linnaeus

In midsummer there is a sudden burst of bright, pink blossoms in Honolulu, due to the flowering of the Crepe Myrtle. Most of these plants are shrubs, but sometimes they attain the size of a small tree. The exceedingly frilled, fringed and crepy petals, five to a flower, occur in such fluffy masses that individual blossoms are often hard to distinguish. There is a white variety and some vary in color to lavender.

Leaves are small, leathery, smooth above, but rough beneath. The stems are brown. The bark, when the plant attains tree size, is so smooth that in some Oriental countries, where it is native, it is called a “monkey slide” tree. ([Plate VIII](#))

A large tree in the Foster Garden bears heads of lavender bloom in summer, making a very striking appearance. This tree is *Lagerstroemia speciosa*.

Crepe myrtles belong to the Henna family and are natives of the Far East.

CROWN FLOWER. GIANT INDIAN MILKWEED

Calotropis gigantea R. Brown

The Crown Flower derives its name from its oddly shaped blossoms which rise in clusters at the branch ends. The flowers are a grayish lavender or a greenish white, the latter being more popular for lei making than the 58 former. Above the five thick, starlike, greenish petals rises a miniature crown which looks as if it had been carved from white jade. It is tipped by the stamens and the five pointed style. These “crowns,” when stripped from the flower, are strung into leis which appear like carved beads; indeed, they have been imitated in carved ivory, perhaps the best of all the imitation flower leis.

The shrub grows rather tall. It is whitish, the stems and thick leaves being covered with down. When cut, the stems give off a milky juice as do other members of the Milkweed family. This plant is a native of India, where, like the *Caesalpinia*, it is sacred to Siva. ([Plate VIII](#))

BEACH NAUPAKA

Scaevola frutescens (Miller) Krause

A quaint little flower, which seems to have been torn in half, since the white petals radiate in only half of a circle, is called Naupaka-kai by the Hawaiians. It is a native plant, found wild on the beaches of these islands, and others in the South Seas. It has been adopted here for seashore planting, since it is resistant to wind and salt spray. There are other species of *Scaevola* growing in Hawaii, some of them preferring a mountain habitat, but all are characterized by the half blossom. The five petals have purple

streaks. Leaves of the Beach Naupaka are thick and grey-green because of a velvety down. They become broader toward the tips than toward the base.

A plant with such an odd flower would be certain to have had legends created about it. They are several in Hawaii, usually on the theme of lovers parted, typified by the incomplete blossom.

The plant belongs to the Goodenia family. It may be seen growing along the seawall of the Halekulani hotel or along the shore lines of the Islands. ([Plate VIII](#))

59

PURPLE, OR TRAILING LANTANA

Lantana sellowiana Link and Otto

A low, covering plant, dotted with small clusters of rosy lavender flowers is the purple Lantana. The flower heads are an inch or so across, each made up of flowerets which are little tubes with five lobes. The foliage is small, stiff and rough, with prominent veins, each leaf minutely scalloped. The plant is woody and firm, by which it can be distinguished from an annual, herbaceous Verbena which is sometimes grown as a ground cover and has flowers of somewhat similar form and color. The Lantana belongs to the Verbena family, accounting for the similarity. The trailing purple Lantana comes from South America. ([Plate VIII](#))

The familiar red and yellow *Lantana camara* is often seen in Hawaii growing wild beside the road. It is an “escape” from gardens. At one time it threatened all island agriculture, for without natural checks in the new territory to which it had been introduced it formed dense thickets which could hardly be eradicated. It was finally controlled by introducing parasites from tropical America, where it is native.

PIKAKE

Jasminum sambac Solander

The Pikake flower is seldom seen growing, for the shrub is rather ungainly, with large stiff paired leaves, and the blossoms make little show. It is when these flowers are strung into leis, however, that they become universally recognized, through their magnificent fragrance. It is regarded by many people as the most enchanting flower scent in the world. The individual flowers are small, waxy, white, usually double. One lei strand is enough to scent a room, but several are usually worn in order to make an attractive showing.

It is this Jasmine that is grown commercially in China and added to tea leaves to make Jasmine tea.

Its peculiar Hawaiian name (pronounced peacocki), was probably derived from association with the white peacocks which used to wander through Ainalahou, the beautiful estate at Waikiki where the Princess Kaiulani lived in the nineties. The lei of Jasmine flowers was the favorite of this young heiress to the Hawaiian throne and by a natural association of favorites, her lei and her birds were called by the same name. ([Plate VIII](#))

CROWN OF THORNS ***Euphorbia splendens* Bojer**

Quantities of very long, sharp thorns on a low shrub bearing small, red flowers is the Crown of Thorns. The little, rosy-scarlet flowers grow in small clusters on longish stems. What appear to be two red petals are really a pair of bracts. The leaves are few, bright green, and appear on the new growth.

The plant, which is usually less than three feet high, forms a dense mass of thorns with its bare, brownish stems. It is sometimes planted in areas from which it is desired to keep people, as in the parking strip before the Advertiser building. The plant is a native of Madagascar, and a member of the Euphorbia family. ([Plate VIII](#))

It is a cousin of the Poinsettia, *Euphorbia pulcherrima* from Mexico, which grows very luxuriously in Hawaii.

SINGAPORE HOLLY

***Malpighia coccigera* Linnaeus**

Miniature, holly-like leaves, crisp, shining and thorny, mark this charming little plant which is not a native of Singapore but of the West Indies. Nor is it a true holly, but a member of the Malpighia family. The plant is covered occasionally with dainty pink flowers, their five petals around the yellow stamens, very fringed and crepy and suggesting in form their cousins, the Orchid vine (*Stigmaphyllon*). They have a slight fragrance. The plant grows rather stiffly and is often seen in pots. ([Plate VIII](#))

61

DOMBEYA

***Dombeya wallichii* Bentham and Hooker**

The Dombeya is a shrub or small tree with large leaves among which hang showy round, drooping heads of many pink flowers. The flower clusters grow at the end of long, downy pedicels and are so heavy that they hang far over. Individual flowers have five pink petals, and the stamens are united into a short tube in the center. Even when these flowers are brown and dried the cluster still hangs on the plant.

The leaves are big and velvety, roundly heart shaped, with lobes. The plant is a native of Madagascar and a member of the Cocoa family. A white flowering shrub of similar appearance is the *Dombeya spectabilis*. A specimen grows in the University of Hawaii grounds. ([Plate IX](#))

KALAMONA

***Cassia glauca* Lamarck**

A very commonly seen shrub or small tree bearing numerous clusters of bright yellow flowers and, at the same time, bunches of brown pods, is the Kalamona. A native of tropical Asia, this plant has become naturalized in

Hawaii and is often seen growing wild, especially in dry places. Its yellow flowers are similar in general form to those of its cousins, the Shower trees, and like them also, the foliage is compound. Each leaf is made up of many medium-sized leaflets. The flowers appear most of the year, but are best in spring and early summer. ([Plate IX](#)) The Hawaiian name has been transferred to this plant from a native Cassia of similar appearance, *C. gaudichaudii*.

HAOLE LEHUA ***Calliandra grandiflora* Benth**

Flowers which are pompons of pink or white stamens, blooming in winter and spring, announce the Haole Lehua. This name, meaning foreign Lehua, is applied also to a closely related species, *Calliandra haematoma* 62 which has similar flowers of a bright pinkish red. Resemblance of these flowers to those of the native Ohia Lehua has resulted in this name being transferred. At the present time they are more commonly seen in leis than the true Lehua. Such leis are particularly beautiful suggesting a garland of marabou feathers. The shrub has small compound leaves and grows twelve feet tall. The flowers are followed by brown seedpods, showing the plant to be a member of the bean family. It is a native of tropical America. ([Plate IX](#))

CUP AND SAUCER PLANT. CHINAMAN'S HAT ***Holmskioldia sanguinea* Retz**

Tall sprays of the *Holmskioldia* are lined with quaintly shaped little flowers, of tawny orange or deep scarlet color. They are well described by the names of Cup and Saucer, or Chinaman's hat. Each is made up of a saucer-shaped bract, which is the most conspicuous part, from the center of which rises a small tubular flower. They bloom the year round and may be found in many gardens.

The shrub is sprawling or half climbing, with small opposite leaves, usually with irregular margins. It is a native of Burma and a member of the Coffee family. ([Plate IX](#))

KONA COFFEE. ARABIAN COFFEE

***Coffea arabica* Linnaeus**

The shrub or small tree that produces the coffee bean of commerce is sometimes grown in Hawaii as an ornamental plant. It is conspicuous for its rich, dark, shining, leaves, strongly veined and for its bright red berries and the fragrance of its small, white flowers. The flowers are starlike, and grow rather inconspicuously in the leaf axils, a few in a cluster. They are followed by the green berries which turn bright red when ripe. These berries usually contain two seeds which are the coffee “beans” of commerce.

The Arabian coffee plant grows best at levels cooler than the average 63 in Honolulu, so the plant is seen on Tantalus and in the high valleys. It is grown commercially in the Kona district, on the Island of Hawaii and the product has been given the name of Kona coffee. Most of the world’s coffee is made from this species, which was originally native to tropical East Africa, but was introduced very early into Arabia. ([Plate IX](#))

Another coffee plant grown in Honolulu is *Coffea liberica*, also a native of tropical Africa. It grows under slightly warmer conditions than the other and so does better in Honolulu. It is larger than *C. arabica*, becoming a small tree. Coffee plants are related to the fragrant Gardenias, and give their name to the family.

CORAL BUSH

***Jatropha multifida* Linnaeus**

Again the similarity of a flower to coral has given the name of Coral Bush to a plant. In the case of the *Jatropha*, the likeness is not far-fetched, for the

Jatropha flower head is very curious and strangely like a small bunch of red coral. Stems and rounded buds are red and glossy. A few flowers open at a time, showing five small petals and yellow stamens. The fruit which follows is a green capsule holding several seeds.

The leaves of the plant are palmate, and deeply divided in seven to eleven slender parts, giving a lacy appearance to the shrub. It grows about ten feet high. ([Plate IX](#))

Flowers of almost identical form grow on a related plant called *Jatropha podagrica*. This plant, however, is small and usually grown in pots. It is characterized by a greatly thickened stalk which seems to be a large bulb rising from the ground. Flowers and a few leaves grow from the top of this stem. The leaves of this plant are not divided, as in the shrub, but are either entire or lobed. Both are natives of Central America, and belong to the Euphorbia family.

LASIANDRA. PLEROMA. PRINCESS FLOWER

Tibouchina semidecandra Cogniaux

Flowers of rich royal purple, a regal and exciting color, make the Lasiandra conspicuous wherever it blooms. It is not commonly seen in Honolulu, because it prefers slightly higher altitudes, but on the road to the Volcano on Hawaii, or at Kokee on Kauai, it has escaped and makes conspicuous purple masses of bloom. The flowers have five velvety petals, and in the center a group of pinkish stamens, which are peculiarly angled.

The leaves are almost as attractive as the flowers being thickly piled with velvety green hairs which create a silver sheen. They are marked laterally by several conspicuous veins. Old leaves scattered over the plant turn bright scarlet and are as noticeable as flowers.

The plant grows as a spreading shrub. It is a native of Brazil and a member of the Melastoma family. ([Plate IX](#))

NATAL PLUM
Carissa grandiflora De Candolle

The Natal plum, (a native of Natal, in South Africa) is characterized by its long sharp thorns, its fragrant white flowers and its bright red fruits which shine conspicuously among the leaves. The plant may attain almost the size of a small tree but is usually smaller and is often used as a hedge. Its thorns make it practically impenetrable.

The flowers have five, waxy, white petals which always twist slightly to the right. They are very fragrant. The red fruits are edible but sub-acid in flavor. The leaves are very glossy and thick, growing opposite each other on the stem. The plant is a member of the Periwinkle family. It may be seen growing along the Makiki side of the Round Top road. ([Plate IX](#))



Plate IX
TROPICAL SHRUBS—CHAPTER IV



- Identification key
(1) *Dombeya*
(2) *Cup and Saucer*
(3) *Kalamona*
(4) *Haole Lehua*
(5) *Coral Bush*
(6) *Lasiandra*
(7) *Kona Coffee*
(8) *Natal Plum*



Plate X
COLORED FOLIAGE SHRUBS—CHAPTER V



Identification key

- (1) *Snow Bush*
- (2) *Assorted Croton Leaves*
- (3) *Purple Eranthemum*
- (4) *Golden Eranthemum*
- (5) *Panax*
- (6) *Spiral Leaved Croton*
- (7) *Beefsteak Plant*
- (8) *Caricature Plant*



Plate XI

FLOWERING VINES—CHAPTER VI



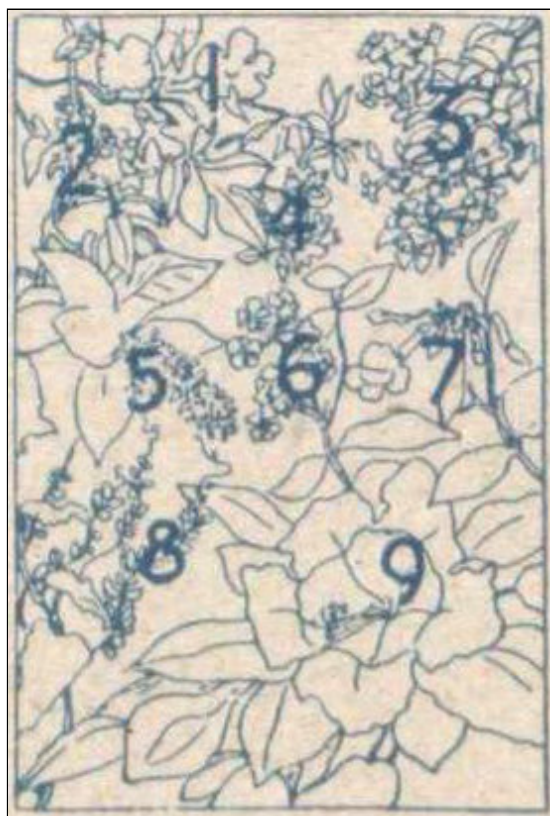
Identification key

- (1) *Yellow Allamanda*
- (2) *Pink Allamanda*
- (3) *Baby Morning-glory*
- (4) *Orange Trumpet Vine*
- (5) *"Mauna-loa"*
- (6) *Pink Bignonia*
- (7) *Wooden Rose*



Plate XII

FLOWERING VINES—CHAPTER VI



- Identification key
- (1) *Cat's Claw Vine*
 - (2) *Galphimia Vine*
 - (3) *Phanera*
 - (4) *Giant Potato Vine*
 - (5) *Sandpaper Vine*
 - (6) *Orchid Vine*
 - (7) *Garlic Vine*
 - (8) *Mexican Creeper*
 - (9) *Cup of Gold*

Chapter V

COLORED FOLIAGE SHRUBS



If Hawaii does not have an autumn season when all the leaves turn red, it has, nevertheless, certain plants which suggest autumn all the year round, with their gorgeously colored foliage. Brilliant tones of red, orange and gold appear perennially in the leaves of many shrubs, while others are more delicately colored in tints of pink, cream and yellow-green. Still others hold very dark shades of maroon, crimson and purple. On most of these plants the flowers are small and inconspicuous, as if the colored leaves took their place in interest.

SNOW BUSH

Phyllanthus nivosus Bull
Variety roseo-pictus

A mass of small, delicate leaves, pale pink and light green in color, on a loose, graceful shrub, is the Snow bush. It is well named, the effect of the

frosty coloring being as if a light fall of snow had touched the leaves. While some plants carry only the light and dark green leaves, others show a rosy coloring in the new growth. This variety is appropriately known as roseo-pictus. The color is strongest in the young parts, the leaves tending to turn to a more even green as they become older. In some, the pink color turns 66 to a dull red. The leaves are rounded in form, about an inch and a half or two inches in length, and grow alternately on the stem. The latter is dark red, with a tendency toward angularity.

Small greenish flowers sometimes hang from long stems in the axils of the leaves, the male and female flowers being separate.

The plant is a native of the South Seas and a member of the Euphorbia family. It is one of Hawaii's most attractive and colorful shrubs being often used as a hedge plant. ([Plate X](#))

CROTON

Codiaeum variegatum Blume

Leading in interest among the colored foliage shrubs in Hawaii is a large group of plants commonly called the Crotons. This name, however, properly belongs to a quite different plant but is used generally by nursery men for this Codiaeum.

Although these Crotons have an almost endless variety of leaf form and color, they all belong to a single species, the difference in appearance being only a matter of horticultural variation. The plants are natives of Malaysia and the Pacific islands, and are members of the Euphorbia family.

To illustrate the wide variations in the Croton leaves, specimens of seven different plants are shown in the upper right hand corner of [Plate X](#) and one more, with spiral leaves, is shown in the lower left hand corner. These eight are perhaps the ones most often seen in Honolulu, but they do not exhaust the local list, and botanical books name many more varieties.

Croton shrubs vary in size, but most of them grow ten or twelve feet tall. Their colors are brightest when growing in full sunlight. Croton leaves remain fresh for some time after they are cut, so that they lend themselves to unusual decorations. The proper name, *Codiaeum*, may have been derived from the Greek word for head, suggesting that the leaves were used to make crowning wreaths. 67

The Croton flowers are small and white, growing in slender racemes in the axils of the leaves. There are separate male and female flowers.

ERANTHEMUM

***Pseuderanthemum atropurpureum* (Bull) Radlkofer (*Erantbemum purpureum*)**

A shrub which might be casually mistaken for one of the Crotons, because of the rich coloring of its leaves, is called *Eranthemum*. The leaf colors are, however, purplish, rose and pink, hues that do not occur in the Crotons. These plants vary greatly among themselves, some having leaves that are mottled in green and white, others with colors that range through the pinkish purples to dark maroon. There is a tendency for the young leaves to have the brightest colors and to turn green as they grow older. The leaves are opposite, strongly veined and rather unevenly margined. The plant belongs to the *Acanthus* family, and is a native of the South Sea islands. ([Plate X](#), 3)

A variety of the purple is the *eldorado*, a horticultural variation, as bright and sunny in its green and gold coloring as the former is dark and rich, deserving its name of *eldorado*, the golden. The pointed leaves are margined and blotched in bright yellow, usually with yellow veins. They grow in opposite pairs and have a tendency to appear in bunches near the ends of the stem. The new leaves are in two tones of yellowish green, the more striking coloration of clear yellow developing as they age. ([Plate X](#), 4)

There is another *Eranthemum*, bearing purple and white flowers, illustrated in [plate VI](#). 68

PANAX
***Polyscias guilfoylei* Bailey**
(Nothopanax guilfoylei, Cogniaux and Merrill)

Visitors to Hawaii are always interested in knowing the name of the commonest hedge plant, a tall slender shrub with grey perpendicular stems and leaves that usually are edged in white or pale green. This is the Panax, a native of the Pacific Islands and a member of the Aralia family.

This shrub is probably one of the most successful hedge plants in the world, since it has few branches and these tend to grow almost straight upward and the foliage is carried right down to the ground. There are several varieties, differing slightly in the form and coloring of leaflets. Some are a flat green, others are edged in white or yellow, or the reverse. All tend to have irregular toothed margins. The leaves are compound, the leaflets opposite, the stems clasping the branch.

Besides the common hedge plant there are a number of Panax varieties in Hawaii, usually grown as specimen or greenhouse plants. One is very fine and dainty, with deeply cut, irregularly shaped leaves. Another is curly and still another is a giant, with leaves eight inches across.

The Panax very rarely flowers. ([Plate X](#))

BEEFSTEAK PLANT
***Acalypha wilkesiana* J. Mueller (of Aargau)**

A plant with bright red foliage, which might easily be taken for one of the Crotons is really an Acalypha, a relative of the striking Chenille plant illustrated in [Plate VIII](#). The leaves of this plant are large and tend to a triangular form. They are basically a bronzy green color, with spreading blotches of pink, red and brown, but the total effect of the plant is one of bright red. These shrubs grow ten feet high and are sometimes used for hedges, being always conspicuous objects on the street. There are a number

of other varieties besides the one with the bright foliage, one having dull rose patches on bronzy leaves.

Insects are attracted to these leaves so that often they are full of holes and sometimes they are reduced to lacy outlines.

Flowers are rather inconspicuous but of two kinds, the male and female. The former appear as small upright spikes of reddish tufts which are the pistils; staminate flowers are brownish and drooping and suggest little rat-tails.

The Acalyphas are members of the Euphorbia family and *A. wilkesiana* is a native of the Pacific Islands. ([Plate X](#))

CARICATURE PLANT. MORADO

Graptophyllum pictum Griffiths

People with good imaginations can see pictures in the yellow or white markings on the green leaves of the Caricature plant. No two leaves are ever quite alike but the “picture” appears always in the center of the leaf rather than along the margins. The leaves are a pointed oval in shape, smooth and rather leathery. They grow in opposite pairs. This plant, too, often is taken casually for one of the Crotons. The shrub will become six or eight feet high.

The flowers are small, tubular and dark red. The original home of the *Graptophyllum* is not known, but it grows widely in the tropics and is popular in India. It belongs to the Acanthus family. ([Plate X](#))

There is another variety with leaves of deep, purplish red and bronze, on which the markings are in a lighter shade.

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OTHER COLORED FOLIAGE PLANTS

In [Chapter VIII](#) will be found described a number of other plants with colored foliage which are not, however, shrubs.

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Chapter VI

FLOWERING VINES



Vines sprawling over rocks and banks, or climbing high over walls and trees to hang out floral banners, make up one of Hawaii's most colorful and interesting floral chapters. While some vines are ever-blooming, most have seasons when they suddenly put on a display of color or of rare beauty, that become, often, the most conspicuous sight of the town. The vines are rather easy to identify on the whole, for there are only a few which resemble each other enough to be confusing.

YELLOW ALLAMANDA

Allamanda henderonii Bulliard

Sprawling green vines, often used as a ground cover, with big yellow flowers every day in the year, are the Allamandas. They are one of the most widely used plants in Hawaii.

There are two yellow species commonly seen, the only essential difference being size. The one with large flowers, about five inches in diameter, is *Allamanda hendersonii*. Its leaves are smooth on both sides. The species with smaller blossoms, about three inches in diameter, is *Allamanda cathartica*. It may be identified, if not in bloom, by the fact that its leaves are somewhat hairy on the under side.

The bright yellow flowers grow in terminal clusters, two or three of the buds opening at one time. The buds are pointed, those of the large A. 72 *hendersonii* being quite brownish in color and looking as if they had been varnished. This brown color blotches the back of the opened flowers. Blossoms are campanulate in form, the tube spreading out into five large, thick, velvety lobes. The throat of the tube in the large flower is also streaked with brown and there are whitish spots at the base of the petals. Flowers of these vines do not last when cut, unless the stem ends have the sticky juice coagulated by holding in very hot water.

The leaves usually appear in fours, forming a cross or whorl where they join the stem. They are a pointed oval in form, thick, smooth and rather light green.

These plants are natives of Guiana and members of the Periwinkle family. There is a fine planting at Vancouver drive and Hunnewell street. ([Plate XI](#))

PINK ALLAMANDA

***Allamanda blachetti* A. De Candolle**

The rose-colored member of the Allamanda group in Hawaii is not so often found as the yellow. The flowers are about the size of the large yellow and in color a deep rose, or almost maroon, with the throat a deeper shade.

The leaves of this species, while they display the whorled growth of the yellow, are smaller and very rough and hairy on both sides.

The vine comes from Brazil. ([Plate XI](#))

BABY MORNING GLORY
***Jacquemontia pentantha* G. Don**
(*Convolvulus mauritanicus*)

Flowers like miniature blue morning-glories, about an inch across, grow on this slender vine. It is charming because of its petite size, and for the beautiful color of the flowers. They have a lighter throat, and white stamens. The buds appear in clusters at the end of the flower stem and one or two flowers open each morning, closing in the afternoon. The leaves are slightly heart-shaped, ending in a sharp point. The stems are reddish. The *Jacquemontia*, which is a native of tropical America, was named after a French botanist, Victor Jacquemont. Formerly it was called *Convolvulus mauritanicus*. ([Plate XI](#))

ORANGE TRUMPET VINE. FIRECRACKER VINE.
HUAPALA
***Bignonia venusta* Ker**

One of the most spectacular events in Hawaii's colorful floral calendar, is the blooming of the Firecracker vine. In late winter, walls of green foliage turn suddenly into a sheet of flaming orange, the masses of flowers seeming like small tongues of fire blazing over the entire vine.

The blossoms grow in end racemes. Each individual flower is a long, slender tube spreading into four or five lobes which curl back against the tube. They often form the outline of a cross, with the fourth lobe split, curiously, into two parts to make the five-part flower. The style and four stamens extend beyond the tube. When the flower begins to fade, the tube slips loose from the calyx; but it is often caught in its fall by the enlarged tip of the style, so that the flower hangs on the vine, to add its color to the mass, for a while longer.

Leaves are glossy and bright green, usually growing as three rather pointed leaflets. Like so many other vines in Hawaii, it is a native of Brazil. The

Hawaiian name, “Huapala” means Sweetheart. ([Plate XI](#))

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“MAUNA LOA”. PUA KA UHI
***Canavalia microcarpa* De Candolle**

Anyone who has remained for long in Hawaii has seen and wondered at the Maunaloa leis, those strangely formal, almost sculptured floral bands which have scale-like, overlapping petals in the center, and are bordered on either edge by rounded projections. The flowers from which these leis are made are a typical pea blossom. Strung together and turned right and left alternately, the “banner” or large top petal is then bent back and held down by being pressed onto the surface of a narrow strip of adhesive tape stretched along the length of the lei. The “keel” of the pea flower forms the border projections.

Originally these leis were made from the Maunaloa flowers, which are botanically *Dioclea altissima*. But these are rarely seen nowadays and most of the Maunaloa leis are made from a closely related flower, the *Canavalia microcarpa*. The blossoms of this vine range in color from white, through orchid pink to lavender and even maroon. They grow in elongated clusters at the tips of the shoots. The stems of the plant are dark red, the leaves are made up of three leaflets, triangular in form, with reddish venetions. This plant is an annual, growing from large, dark roundish seeds. It is a native of Brazil and grows wild in Hawaii. ([Plate XI](#))

The true Maunaloa is very similar to it in general form.

PINK BIGNONIA
***Pandorea jasminoides* Schumann**

There are several kinds of vines growing in Honolulu which have clusters of pink or orchid colored trumpet shaped flowers, often with a dark red throat. These are usually called vaguely Pink Bignonia, for they are either

members of the Bignonia family or closely related. Their botanical relationships are not easily straightened out for the layman, since all are rather similar in appearance. On [Plate XI](#) is shown *Pandorea jasminoides*, a vine from Australia. Others are *Bignonia jasminoides* and *Bignonia regina* from tropical America. All are attractive with their pinkish bell-shaped flowers and fine green foliage. *Pandorea jasminoides* may be seen growing on a wall on the lower part of Diamond Head Road. 75

WOODEN ROSE

Ipomoea tuberosa Linnaeus

One of the strangest and most attractive of Hawaii's plant novelties is the "Wooden Rose," which looks indeed like some wonderful bit of carving, rubbed to an exquisite satiny brown finish. The "rose" however, is really the dried seed pod of a species of morning-glory, as anyone familiar with the ordinary morning-glory seed will at once recognize. The central ball holds the seeds while the enlarged, dried calyx which surrounds it, appears to be petals.

In Hawaii, the vine is a perennial, grown from seeds. Its strong shoots spread rampantly during the summer month, climbing high into trees or covering buildings and fences. The leaf is divided into seven pointed lobes. The flowers first appear in autumn. They are yellow, small, rather inconspicuous and tubular, like the small yellow morning-glory which they really are. After they fall, the calyx begins to develop until it has enlarged into what looks like an immense pointed, cream-colored bud. As this begins to dry, it opens, showing the enlarged seed case. In a few days, the "wood rose" is stiff and brown. About three months are required from the time the blossom appears until the seed pod is ready to cut. These pods may be used as a long-lived decoration; and, since the flowers grow at intervals along the shoot in the leaf axils, graceful lengths of stem with many roses can be used for flower arrangements. 76

The vine grows generally throughout the tropics but is sometimes called Ceylon Morning-glory. ([Plate XI](#))

CAT'S CLAW VINE. HUG-ME-TIGHT

Bignonia unguis-cati Linnaeus

The three-pointed, claw-like tendrils by which this vine clings closely to trees, or walls, have given it the two names by which it is commonly known. But it will be readily recognized and remembered from the cloth of gold it flings several times a year over everything it covers. Individual flowers are trumpet shaped, with five spreading lobes, about two inches across. The color is a clear, canary yellow. ([Plate XII](#))

The leaves are compound, the paired leaflets being pointed and narrow. The plant is a native of tropical America where it is related to some of the giant lianas that creep through the Brazilian jungle. Its most conspicuous relative in Honolulu is the Firecracker vine, *Bignonia venusta*, illustrated on [Plate XI](#).

GALPHIMIA VINE

Tristellateia australis A. Richard

This yellow flowering vine is rather rare as yet in Honolulu, but is bound to grow in popularity as its attractive flowers and leaves become known. The color of the leaves is a light yellow-green. They are opposite, smooth, thick and waxy, with a tendency to fold along the mid-rib.

The flowers appear on pendant end-shoots, in long clusters. They have five, pale, yellow petals and in the center a group of short, red stamens. 77 Probably this vine can be most readily identified by this touch of red in the middle of the yellow blossom. It belongs to the Malpighia family and hence is a cousin of the popular Galphimia shrub. It is a native of Australasia. ([Plate XII](#))

GIANT POTATO VINE

***Solanum wendlandii* Hooker**

Delicate, pale, periwinkle-blue flowers appear in large, loose, clusters in early summer and again in autumn on the Giant Potato vine. The petals of the flowers are not separate, but are connected, curving outward slightly, making the flower almost pentagonal in outline. The mid-rib of each petal is of slightly different texture and lighter color than the rest of the corolla. In the center of the flower is a group of thick stamens forming a low column.

The leaves of this vine are not all of one shape, but vary in form from a simple outline to one that is deeply lobed, with the end lobe sometimes larger than the others. The leaves are smooth in texture but have occasional prickles. ([Plate XII](#))

Another “Potato vine” in Hawaii is *Solanum seaforthianum*. It has small flowers of a rich purple-blue, about an inch across, which appear in loose clusters in summer. They are of the same pentagonal form as the larger ones and have a bright yellow center created by the stamen.

The foliage of *S. seaforthianum* is small; but, like the large potato vine, varies in form.

The giant vine is a native of Costa Rica while the smaller one came from Brazil. Both belong to the Solanum, or nightshade family.

A close relative is the Giant Potato Tree which has flowers of very similar form and color. See [Plate II](#).

PHANERA

***Bauhinia corymbosa* Roxburgh**

The Phanera carries large, loose, corymbose clusters of small pale, pinkish flowers, during the summer months. The flowers are about an inch across and have five delicately fluted white petals. These may be flushed with

pink. Several long bright red stamens project from the center and give the flower cluster a pinkish effect.

The leaves seem to be paired, but are really deeply lobed, their outer edges rounded, the notch cut in deeply. The nerves are almost parallel. This peculiar leaf shows the relationship of this vine to other members of the same family, especially the Orchid tree and the St. Thomas tree. The genus was named for the twin Bauhin brothers who were herbalists in the 16th century.

The flowers are followed by long, flat, purplish-brown pods, showing this plant belongs to the legume family. Its native home is China. ([Plate XII](#))

SANDPAPER VINE. PURPLE WREATH

***Petrea volubilis* Linnaeus**

One of the most exciting experiences in Hawaii is to come upon a plant of the Petrea in full bloom. The cascading racemes of lavender-blue flowers cover the plant completely, turning it into a tumbling fall of lacy blue. The calyx seems like a flower in itself, being starlike, five pointed and periwinkle blue. The true flower is a rich violet in color and looks something like a real violet growing in the center of the calyx. This true blossom falls off the plant in a day or so, leaving the calyxes to suggest a cluster of Wistaria blossoms. Each raceme is seven or eight inches long and carries fifteen to thirty flowers. The plant blooms several times during the year, at least once in spring and again in summer. 79

The leaves are yellowish or grey-green and very rough in feeling. They give the plant its name of Sandpaper vine. It is a native of Brazil and a member of the Verbena family. A specimen may be seen on Metcalf street, near Hunnewell. ([Plate XII](#))

ORCHID VINE

***Stigmaphyllon littorale* A. Jussieu**

Clusters of delicate, yellow flowers, suggesting small yellow orchids have given the name of Orchid to the two *Stigmaphyllons* which grow in Hawaii. They are, however, in no way related to Orchids but belong to the Malpighia family. In recent years they have become very popular in Honolulu but cannot yet be found widespread in gardens.

Individual blossoms have five unequal petals of a crepy, satiny, texture and a clear bright yellow color. The flower illustrated in [Plate XII](#) is *Stigmaphyllon littorale*. Its flowers are smaller and more numerous than the cousin, which is *Stigmaphyllon ciliatum*, but the form of the two flowers is very much alike. The foliage of the two plants, however, is different for while both have strong, leathery, shining leaves, those of *S. littorale* are oval, while those of *S. ciliatum* are small, pointed, quaintly heart-shaped and as the botanist says, they are “ciliate,” that is, fringed by coarse hairs. From this is derived its specific name of *ciliatum*.

These two vines are natives of tropical America.

GARLIC VINE

***Cydistia aequinoctialis* Miers**

A vine with charming clusters of orchid-colored, bell-shaped flowers, radiates a most disagreeable odor of bad garlic, which gives it the 80 inevitable name of Garlic Vine, or, since it is a species of the widespread Bignonia family, the name of garlic-scented Bignonia. The flowers appear most prolifically in autumn and spring, but a few may be found almost any time. The white-throated tube of the blossom is slightly flattened and then broadens into five lobes of a purplish-orchid color. At the bottom of the tube are yellow stamens.

The leaves are a rich, glossy green, growing in opposite pairs, so that four appear to grow from one point on the stem. A straight tendril extends from between the pairs near the end of the branch. ([Plate XII](#))

MEXICAN CREEPER. CHAIN OF LOVE

Antigonon leptopus Hooker and Arnott

Lace-like masses of small, bright-pink flowers clambering by curling tendrils over weeds, rocks or trees, announce the Mexican creeper. Sometimes the white variety is seen and there are also pale pink hybrids. In its native Latin America, this plant is called *Cadena del Amor*, or Chain of Love, since the flowers suggest a string of small pink hearts. The Mexicans have also given it other sentimental names such as *Rosa de Montana*, *Corallita* and *San Miguelito*.

The flower chains branch in a rather angular way giving an effect that is peculiarly picturesque. They lend themselves to flower arrangements of special charm. For this purpose the white variety is often more useful than the bright pink, since it blends better with the average interior color scheme. The only drawback is that the flowers fall rather quickly, but they are worth arranging even for a short time.

The leaves are heart-shaped with wavy margins. There are no petals, the colored portion of the flower being the calyx, with five petal-like sepals. The seeds form and remain inside the dried calyx. The plant belongs to the buckwheat family. ([Plate XII](#))



Plate XIII

FLOWERING VINES—CHAPTER VI



Identification key

- (1) *Bleeding Heart*
- (2) *Kuhio Vine*
- (3) *Porana*
- (4) *Crimson Lake Bougainvillea*
- (5) *White Thunbergia*
- (6) *Wax Vine*
- (7) *Beaumontia*
- (8) *Blue Butterfly Pea*



Plate XIV
GINGER BLOSSOMS—CHAPTER VII



- Identification key
(1) *Shell Ginger*
(2) *Yellow Ginger*
(3) *Crepe Ginger*
(4) *Red Ginger*
(5) *Kahili Ginger*
(6) *White Ginger*
(7) *Torch Ginger*



Plate XV
SPECIAL TROPICAL FLOWERS—CHAPTER VIII



Identification key

- (1) *Spider Lily*
- (2) *White Bird of Paradise*
- (3) *Bird of Paradise*
- (4) *Golden Heliconia*
- (5) *Lobster Claw*
- (6) *White Anthurium*
- (7) *Red Anthurium*
- (8) *Flowering Banana*
- (9) *Spathiphyllum*



Plate XVI
SPECIAL TROPICAL FLOWERS—CHAPTER VIII



- Identification key
(1) *Dieffenbachia*
(2) *Green Ti*
(3) *Pothos*
(4) *Red Ti*
(5) *Caladium*
(6) *Monstera*
(7) *Rhoen*
(8) *A'pe*

CUP OF GOLD
Solandra guttata Don

One of the most magnificent flowers in Hawaii is the great Cup of Gold blossom. It could be more appropriately called a golden chalice than a mere cup, for the blossom is nine inches long above its stem-like tube, and wide and curving in outline. It is the rich golden color of a ripe banana, and brownish streaks on the petals increase this suggestion. Its fragrance, however, is the deep, heady scent of ripe apricots. The huge buds, waxy in texture, when they once start to unfold, move so rapidly that the backward

curving movement may be easily observed. The plant blooms in the winter and spring months. Its leaves are large and rather pointed. ([Plate XII](#))

There is a very similar flower which is cream-white in color, hence called the Silver Cup. This is *Solandra grandiflora*. These two are members of the Potato family. They are natives of Mexico and tropical America, where they are called in Spanish, *Copa de Oro*.

BLEEDING HEART. BAG FLOWER ***Clerodendron thomsonae Balfour***

The quaint little red and white flowers of this vine appear in clusters during the winter and spring months. The vine is usually rather small and is often grown in pots. The crimson portion is the true flower, while the “heart” or “bag” is the white calyx. The red flower is composed of a slender tube extending beyond the calyx and spreading into five lobes. A group of 82 fine stamens protrudes beyond the flower. The leaves are opposite, oblong-ovate, and slightly rough to the touch.

This *Clerodendron*, which is a member of the Verbenaceae family, is a native of West Africa. ([Plate XIII](#))

KUHIO VINE. PRINCE'S VINE ***Ipomoea horsfalliae Hooker***

A close covering mass of magenta-crimson flowers in autumn, winter or spring, is almost sure to be the Kuhio Vine. (The Crimson Lake Bougainvillea, though of about the same color, hangs in long swaying sprays.) The Kuhio vine, one of the morning-glories, is a native of India and is found growing widely in the tropics. It was brought to Hawaii by Prince Kuhio when he was the Territory's delegate in Washington. For years, a large vine grew over his house at Waikiki, on that portion of the beach now

known as Kuhio Park. It is natural that the plant should have been called the Prince's vine, or Kuhio vine.

Individual flowers are shaped like a long bell with a waxy tube and a wide mouth, made up of five lobes. The leaves are a dark, rich green and divided, usually, into five parts. ([Plate XIII](#))

PORANA VINE

Porana paniculata Roxburgh

A mass of tiny white flowers, so small and so numerous they suggest a drift of smoke, or a light fall of snow, is the Porana vine in bloom. The flowering period is late summer and autumn. The rest of the year the plant carries its thick, grey, felt-like leaves along walls and trellises. The leaves are opposite, either heart-shaped or oval, and rather large.

Individual flowers are shaped like minute white morning-glories, the Porana being a member of this family. The tiny white blossoms appear in huge lacy panicles at the end of the branches. They can be used as cut flowers for a short period before the blossoms begin to fall, and they are popular for use in bridal bouquets in Hawaii.

The Porana is a native of India and Malaya, where it grows to a great height in the jungles. Its name is said to be derived from the native Javanese name. ([Plate XIII](#))

CRIMSON LAKE BOUGAINVILLEA

Bougainvillea glabra, var. Sanderiana Choisy

Long, waving sprays of bright crimson flowers are a conspicuous feature of Honolulu gardens in winter, spring, and early summer. These sprays grow on the Crimson Lake Bougainvillea, a close cousin of the purple flowering species which is so familiar in California and other temperate areas. The

purple forms grow in Hawaii, also. One of them, *Bougainvillea spectabilis*, is a mass of purple in the spring; its smaller, ever-blooming form is the variety *parviflora*. This same group includes, also, the orange and tawny-hued form, which is *B. spectabilis*, variety *lateritia*.

Flower shades of all these plants vary considerably, the purple hues ranging through lavender and pink, while the entire color range in golden tones appears in the *lateritia*, varying from a golden buff through rich terra cotta red, to orange and almost scarlet. All are practically alike in form. The brilliant color is not due to the true flower but to modified leaves or bracts, three of which enclose the true flowers. The latter are small, tubular and pale yellow. The leaves of the plant are small, rather triangular in shape, with wavy margins.

The stems of Crimson Lake have large thorns. The plant climbs strongly during the summer months, then in winter, its clusters of flowers appear at the end of its branches. ([Plate XIII](#))

The orange or terracotta-colored variety makes a gorgeous mass of flaming color in the winter season while the purples add their exotic hue to the kaleidoscope. The grounds of St. Louis College make a feature of purple and red *Bougainvillea*, while Punahou School has a fine plant of the orange.

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Bougainvilleas are natives of Brazil. They were named for de Bougainville, a French navigator who lived from 1729 to 1811. The plant belongs to the four o'clock family.

WHITE THUNBERGIA

Thunbergia grandiflora Roxburgh Var. *alba*.

One of the most conspicuous flowers in Hawaii carries starry white flowers about four inches across against its green wall of leaves, or dramatically drops these flowers in waving streamers sometimes two or three feet long.

Rows of buds develop at the branch ends and the flowers begin to open at the top. As they open, the branch grows also until it nearly doubles its first length.

The individual flowers are funnel-shaped with a pale yellow throat, the tube broadening to five lobes. Leaves are roughly oval, or shaped like an angular heart, and are quite rough to the touch. Because of the dramatic appearance of the long white streamers, the plant has become very popular in Honolulu in recent years. It is a native of India, a member of the Acanthus family. ([Plate XIII](#))

The blue flowering species, *Thunbergia laurifolia* was established much earlier than the white variety. The latter however has outstripped the former in popularity. The blue Thunbergia does not trail its flowers so conspicuously as does the white and has leaves which suggest the laurel, giving the specific name, *laurifolia* to the plant. It is sometimes called Blue Sky Flower.

WAX VINE

Hoya carnosa R. Brown

Noticeable for its thick, shining, oval leaves is the Wax Vine. Hidden among the leaves are the clusters of fragrant, waxy, white flowers. They grow in umbels, the flower stems radiating from a single point on the main stem. The small blossoms are shaped like creamy-white stars, and each flower contains a smaller star in its center. This is white against a pink flush at the base of the petals. In another variety, the flower is brownish. They give off a strong fragrance, especially in the evening. ([Plate XIII](#))

BEAUMONTIA VINE. NEPAL TRUMPET FLOWER

Beaumontia grandiflora Wallich

Immense clusters of large, striking, white flowers seen on a strong rampant vine, mean that the *Beaumontia* is in bloom. The season is winter and spring. The flowers are about six inches across, papery in texture and a dead white in color, except for a pink flush on the back, and pale green in the center. They are cup shaped, with five wavy lobes. In the center of the flower rise the five stamens, white and pale green in color and joined at the tip into a point. The flower has a delicate fragrance, matching its fragile appearance. If cut early in the morning and plunged deeply into water for awhile, it will be successful as a cut flower. The blossoms are often used in wedding bouquets in Hawaii.

The vine grows to a large size with long, large leaves prominently veined and a shining bright green in color. The plant is a native of tropical Asia. It is a member of the Periwinkle family. ([Plate XIII](#))

BLUE BUTTERFLY PEA

Clitoria ternatea Linnaeus

Blossoms of a true cerulean blue are exceedingly rare in the flower world, but those of the Butterfly Pea are of this hue. Though small and scattered on the vine, these little flowers are delightful for their 86 gorgeous color and unusual shape. As members of the pea family, they are shaped like a modified pea blossom, the “banner” or large back petal being oval, the wings very small. The banner usually has a white mark on the base. Sometimes the flowers occur double and there is also a white variety.

The foliage is compound, the leaflets being rounded. The plant, which is an annual in colder climates, grows rather thickly. The dried pea-like pods which follow the flowers hang on the vine a long time. The seeds grow easily.

The plant gets its name from the island of Ternate in the East Indies, but is considered a cosmopolitan in the tropics. ([Plate XIII](#))

Chapter VII

GINGER BLOSSOMS



Leading among Hawaii's special flowers are those of the Ginger family. They are usually exotic in form, colorful, and often intoxicatingly fragrant. The name, Ginger, covers several groups or genera, which vary considerably in appearance although the botanist can distinguish the similarities which relate them. Gingers are not far removed from the Cannas and Bananas; hence, they are reedlike plants, with fibrous stalks and blade-shaped leaves. Some are short, hardly more than a ground cover, others grow twelve or fifteen feet in height.

A native ginger, called Awapuhi by the Hawaiians, and *Zingiber zerumbet*, by the botanist, grows in the Hawaiian forests. Its leaves form a ground cover a foot or two high. In spring the flower heads spring up, bulbous and reddish, composed of scaly bracts out of which appear the small, inconspicuous, yellowish flowers.

The plant from whose root is made the dried ginger of gingerbread also grows in Hawaii. It is called Chinese Ginger or *Zingiber officinalis*. From

its light-skinned rhizome is made the Chinese candied and preserved ginger, and bits of the fresh root, or the young shoot, often add piquancy to Chinese cooking.

SHELL GINGER. PINK PORCELAIN GINGER

Alpinia nutans Roxburgh

Like a strand of closely strung shells, the buds of the Shell Ginger droop gracefully from the ends of the stalks. Each bud is thin and porcelain-like, white, pointed and tipped with bright pink. These shell-like buds open, a few at a time, and the flower pushes out. It has thin, white petals while a larger, ruffled portion is yellow, marked with red vein-like lines. One of the stamens also has a petal-like development. The fruit is a yellow ball.

The plant is made up of luxuriant stalks of long-bladed leaves which grow five to twelve feet high. It is a native of the East Indies. ([Plate XIV](#))

Another ginger of this genus, *Alpinia mutica*, is conspicuous in Hawaii not so much for its flowers, which are also yellow and white, as for its bright, orange-colored fruit—like round balls. These remain on the plant a long time and make good cut decorations.

YELLOW GINGER

Hedychium flavum Roxburgh

The Yellow Ginger has flowers like slender moths of pale, creamy, yellow. They rise at the end of narrow tubes above a green head composed of scaly bracts. One blossom emerges from behind each scale and the buds of those above it peep out like yellow quills. The flower has three petals, two paired and wing-like, the third large and looking like a second pair of wings, folded together. There are three slender sepals and a long filament of deeper color, holding the pistil and stamen. Yellow Ginger blossoms have a delicate fragrance, delightful when perfectly fresh, a little rank when the

least bit wilted. Leis made before the buds open, have the smooth 89 quality of old ivory carvings.

The plant has characteristic canes of long leaves which grow five to eight feet. It prefers cool locations, growing wild along the Nuuanu Pali road. Yellow Ginger is a native of India. ([Plate XIV](#))

CREPE GINGER. COSTUS

Costus speciosus Smith

(*Costus spicatus*^[2])

Ruffled and fringed white flowers of odd form emerge, two or three at a time, from behind the scales of the large, brownish-red bracts of the costus. These form a dark head, often so large as to suggest a pineapple. The white flowers have a curious structure. The three, true petals are white and rather inconspicuous behind a large, crepy, white portion which seems to be the petal but is really a greatly modified stamen, called a staminoidium. This rolls into a bell form, with fringed and fluted edges and a pale yellow throat. A second modified stamen carries the anthers and has a yellow tip, making it appear like the usual center of a flower. The stems of this plant have a tendency to curve spirally. The leaves are not so long and blade-like as in other gingers and are arranged spirally on the stem. The plant is a native of the East Indies. ([Plate XIV](#))

RED GINGER. OSTRICH PLUME GINGER

Alpinia purpurata (Vieillard) Schumann

Long rosy red heads among the green leaves are sufficiently suggestive of ostrich plumes to justify this name for the Red Flowering Ginger. The head is made up of large, thin, petal-like bracts and is the conspicuous 90 portion. The true flowers are small and whitish and appear occasionally from behind the bracts. A curious characteristic of this plant is

that adventitious plantlets form in the head. These grow easily when planted.

Red Ginger is a native of Malaya. ([Plate XIV](#))

KAHILI GINGER

Hedychium gardnerianum Roscoe

The local name for this ginger is derived from the *kahili*, an item that was part of the regalia of early Hawaiian chieftains. A *kahili* was made from a pole or wand, near the top of which, and at right angles to it, were affixed long wing or tail feathers from certain large birds, forming a cylindrical head. This was carried, like a banner, wherever the chief went, to announce his rank and presence.

The blossoming head of the Ginger called after the *kahili* shows an obvious resemblance. The small yellow flowers on long, stem-like tubes form a cylinder around the top of the stalk, while the resemblance to feathers is enhanced by long, red, filaments which are very striking against the yellow of the petals. Individual flowers have the general form of the Yellow Ginger, but are much smaller and their color is not creamy, but bright yellow. The flower stalks may be six feet long and rise above the rest of the plant. This species is native to the lower Himalayan region. ([Plate XIV](#))

WHITE GINGER. GINGER LILY

Hedychium coronarium Koenig

Most romantic of all the Gingers, because of its white, ethereal delicacy and enchanting fragrance, the White Ginger blossom is larger and fuller 91 than the yellow, but has the same moth-like form. The petals, however, hold a shimmering, almost crystalline moon-whiteness which seems unearthly. The slender filament rises in the center like an insect antenna. The flowers are lifted in snowy clusters above the lush green of

their long leaves, each flower head centered by a smooth, waxen, green bulb made up of the scale-like bracts. Behind each bract a flower bud pushes out. Just before they open these buds are strung into leis which are one of the favorites in the Islands.

The plant will grow to eight feet if the soil is moist. It is a native of tropical Asia. ([Plate XIV](#))

TORCH GINGER
Phaeomeria magnifica (Roscoe) Schumann
(Phaeomeria speciosa)

If the White Ginger is the most romantic of this group of plants, the Torch Ginger is the most magnificent and spectacular. The plant is a clump of tall bamboo-like stalks, fifteen feet high, carrying large leaf blades. There are two varieties red and pink, the one with red flowers having bronzy leaves, while the pink has bright green leaves. Under this clump, in spring, seeming almost like an independent plant, pushes up the large flower stalk. It grows from three to six feet tall and carries no leaves, but at the end develops the head which is one of the most showy things in the flower world.

It is a waxen cone made up of innumerable bracts, pink or red, around which is a frill-like involucre of the same colors. The head is most attractive before the small, inconspicuous flowers begin to appear from behind the bracts, making them rather ragged. The general form of the flower head suggests a formalized torch. The flowers lend themselves to arrangements that can be almost monumental.

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Torch ginger is a native of the Netherlands East Indies. ([Plate XIV](#))

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Chapter VIII

SPECIAL TROPICAL FLOWERS



Many plants grow out of doors in Hawaii which are only seen in greenhouses in cooler climates. These include Orchids, which often make purple cascades from baskets hanging on trees, and other kinds which grow in the ground. In Hawaii, however, as in other places, the finer collections of Orchids are grown in greenhouses. This is not for warmth, since the walls of these houses are partly of wire screening, but to protect the plants from rain, wind and insects. The best plants of the island Orchid collections are usually displayed twice a year, at spring and autumn shows in the Honolulu Academy of Arts, where anyone interested may view them.

Plants which appear particularly tropical and exotic are those with large, lush, leaves and strange colorful flowers. Such plants do not require growing conditions any more tropical than do other things listed in previous chapters, but they look as if they did and are here grouped together. The ones selected for description do not exhaust the list by any means, but they are, perhaps, the ones most frequently seen.

A good collection of tropical exotics grows in the greenhouse at the Foster Gardens, a city park open to the public.

SPIDER LILIES

Crinum species

In Hawaii the name of Spider Lily is given to a number of liliaceous plants which have similar flowers, that is, with six, thin, spidery petals and six stamens. By a stretch of the imagination these flowers might be thought of as giant white spiders. The botany of these lilies is much confused and the local ones have never been satisfactorily straightened out. But there are at least three groups covered by the popular name, the chief one being *Crinum*. Others are *Hymenocallis* and *Pancratium*. All are members of the *Amaryllis* family. 94

These plants have bulbous roots which send up a clump of long, blade-like leaves. They vary from one or two feet in length to giants four to six feet long. The flowers are usually white, although sometimes tinged with dark red, and sometimes they have red stamens and stems. Many of these flowers are very fragrant. The Spider lilies are one of the staples of a Hawaiian garden. ([Plate XV](#))

THE BANANA FAMILY

Plants related to the Banana or *Musa* family, supply some of the most exotically shaped and colored flowers in Hawaii. The fruiting Banana does not have conspicuous flowers, but it grows as a graceful tree. The flowering stalk holds large dark red bracts under which are the small yellow tubular flowers. These point upward, as do the fruits into which they develop. The man in the fruit store hangs the bunch upside down.

A relative of the Banana which often attracts attention is the Traveller's Palm, *Ravenala madagascariensis*, which is not, of course, a palm any

more than a Banana is a palm. The Traveller's Palm has the large leaves of the Banana, but they are arranged in one plane, like the sticks of a giant fan. Some other members of the Banana family are included in [Plate XV](#).

WHITE BIRD OF PARADISE

Strelitzia nicolai Thunberg

One of the most curious flowers in Hawaii is the White Bird of Paradise, so called, no doubt, because of its resemblance to its relative, the blue and orange colored Bird of Paradise. The resemblance, however, is not close enough for the white really to look like a bird, as does the orange. The white flowers grow out of a large boat-shaped sheath or keel, deep purplish grey in color, of which there are often two or three in a cluster. The flowers break out of the top of this sheath, one at a time, like white sails. There are three petals with a pale blue staminodium. The keel frequently is smeared with a gummy substance which must be removed before the flower becomes attractive for decoration.

The plant on which they grow is a small tree, with Banana-like leaves, arranged in several small fans on the order of the Traveller's Palm. It is a member of the Banana family and a native of South Africa. ([Plate XV](#))

BIRD OF PARADISE

Strelitzia reginae Banks

The long stalk of this flower looks like the neck of a bird holding a head with long beak and a gorgeous crest. The "head" is a pointed sheath, greyish in color, and the crest of the bird is made up of the flowers lifting out of this sheath. There are about six of them in the sheath and since one pushes out every day or so, the cluster becomes larger and more colorful as it becomes older. Each flower has three pointed petals, brilliantly orange in color, and a blue staminodium shaped like an arrow head. The effect is unusual and exotic in the extreme. The flower is scentless.

The flower stalks grow slightly above the clump of stiff leaves which compose the plant. The leaves, which may be three or four feet long, are paddle shaped and heavy, their edges curving together. This plant, too, is a relative of the Banana and a native of South Africa.

In arranging Bird of Paradise flowers, an effective way is to place them with all the “heads” turned in one direction, which gives the suggestion of a flock of birds in flight. ([Plate XV](#))

GOLDEN HELICONIA

Heliconia latispatha Benth

Other relatives of the Banana are the Heliconias of which there are a number growing in Hawaii. The plant is made up of a clump of tall, paddle-shaped leaves, often ten feet high. The flowers of most species grow below the leaves, but the Golden Heliconia flower pushes above them.

The inflorescence consists of a series of narrow, pointed keels, a deep golden yellow in color. The real flowers are inside these sheaths, inconspicuous and hardly noticeable.

This plant is a native of tropical America. ([Plate XV](#))

LOBSTER CLAW

Heliconia humilis Jacquin

Generally similar to the Golden Heliconia, the keels of the Lobster Claw are much thicker and closer together, and arranged on opposite sides of the stem, in one plane. They are the brilliant red of a boiled lobster and the general form of the keel suggests the claws of the creature. The inconspicuous flowers are inside. As the keels hold rain water, the flowers often start to decay while the sheaths are still bright and fresh, giving a sour, disagreeable smell to the stalk, until it has been thoroughly washed. This

done, they last a long time as decorations. The plant is a clump of tall leaves. It is a native of tropical America. ([Plate XV](#))

There are a number of other Heliconias, one of the most commonly seen having sheaths that are pinkish, edged with yellow and green. This is *Heliconia elongata*.

ANTHURIUM. FLAMINGO FLOWER

Anthurium andraeanum Linden

Among the most popular of Hawaii's exotic flowers are the Anthuriums, for the very good reason that they will last as long as three weeks if they are cut in their prime. They are, besides, large and exquisitely waxen, ranging 97 in color from pure white, through all shades of pink to deep, rich red.

They belong to the Arum family, of which the Calla lily is also a member, and the Anthurium blossom is similar in general form to the Calla. That is, it possesses a large, heart-shaped bract, called the spathe, which is thick and waxen, almost artificial in appearance. From this spathe rises a column, called the spadix, which may be white, pinkish or yellow in color. Packed tightly together on this column are the true flowers, usually so small they are hardly noticeable. When fertilized they may develop small berries with seeds, which grow readily.

The leaves are heart shaped and rather long stemmed and spring from a central stalk. Usually they are grown in pots but sometimes are seen in the ground. ([Plate XV](#))

This Anthurium is a native of Colombo.

There are a number of other Anthuriums, but none with such fine blossoms grows in Hawaii. The others here are foliage plants with beautiful, large, velvety leaves, decoratively veined.

PINK OR PURPLE FLOWERING BANANA

Musa rosacea Jacquin

Hawaii has one species of Banana which is grown for its flowers alone, since the small fruits it bears are not edible. This is the Purple Flowering Banana which consists of a pointed head of rosy orchid-colored bracts. These bracts fall open two or three at a time to reveal the small, upstanding, tubular, yellow flowers which grow in “hands” part way around the stalk. As in the fruiting Banana, the flowers nearer the base are female, while those at the tip are male. The flower is one of the most showy and curious of all Hawaii’s blossoms.

It grows on a small plant, of typical Banana form, about eight feet high, with large, lush leaves. This species is a native of India. ([Plate XV](#))

In handling this flower care should be taken not to let the cut stalk touch clothing, as the juice leaves a permanent stain.

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SPATHIPHYLLUM

Spathiphyllum species

A blossom like a small, white Anthurium, but more fragile and with a large, rough spadix, is the Spathiphyllum. The leaves of this plant grow about two feet high and are long, pointed, blade-like and a very rich, dark green. The plant is much used as a low-growing cover in shady tropical gardens, and in pots. Unfortunately the exquisite white flowers do not last long when cut, a day being about all that can be expected.

The species commonly grown around Honolulu is called *Spathiphyllum clevelandii*, but this is a horticultural name and does not appear in botanical literature. The local species has not yet been satisfactorily determined. The plant is a member of the Arum family and a native of tropical America. ([Plate XV](#))

DIEFFENBACHIA

Dieffenbachia seguine Schott

Frequently seen in collections of tropical plants, on porches or in greenhouses, and sometimes growing in the ground, are the large, green and white leaves of the Dieffenbachias. The species was named for J. F. Dieffenbach, a German botanist of the last century. There are a number of varieties, which differ as to the shape of the leaf blade and the pattern of white on them. One of these is shown on the plate, the one most frequently grown in Honolulu. The flowers are small and seldom seen. They are made up of the spathe and spadix characteristic of the Arum family.

The plant grows as a thick stalk which often lies on the ground for a distance before lifting its head of leaves. The leaf stems clasp this stalk and rings are left when they fall off. The Dieffenbachias are natives of central and South America. ([Plate XVI](#))

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GREEN TI

Cordyline terminalis Kunth

A plant which grows wild and very abundantly in the lower, wet forests of Hawaii is called Ti by the Hawaiians, (pronounced tea). It is primarily a leaf plant, the leaf blades being two or three feet long and very glossy, thick, and strong in texture. They do not wilt easily and so are useful for many things. Shredded to the midrib and strung together, they form the green skirt of the modern hula dancer. At native feasts they are used to cover the table instead of a cloth and sections of the leaves serve as plates. In some meat markets, sections of Ti leaf are used instead of waxed paper to wrap up meat, while leis wrapped in a bundle of leaves remain fresh for a long time. For culinary purposes, pieces of fish and pork, along with young leaves of the taro which taste like spinach, are wrapped in Ti leaves and steamed. The resulting dish is called a lau-lau. The Ti leaf imparts a characteristic flavor to the food it enwraps.

The Ti plant grows as a tall stalk, often woody at the base, which may be twelve feet high. The leaves appear in a tuft at the top of the stalk. From among these leaves in winter and spring comes the flower cluster. The blossoms are very small and tubular, really minute lilies, a creamy white in color, with sepals that are mauve or pinkish, so that the effect is a mass of purplish or pinkish grey and cream. The flower cluster is much branched and very graceful, being attractive even when dried. The flowers on the plant sometimes develop seed. The plant is much used in gardens for hedges and for background plantings. ([Plate XVI](#))

A section of the old woody stem, several inches in diameter and several inches long, if placed in a shallow bowl of water will develop new shoots and become an attractive house plant.

The Ti grows on the South Sea islands, in India and Southern China. It is a member of the lily family. It enters many Polynesian myths and stories, for it was widely used by the ancient islanders as well as by their modern descendants. The thick root stock was used by early white men in Hawaii to make *okolehao*, the starchy root being boiled, fermented and distilled. 100

RED TI ***Cordyline terminalis***

The Ti plant varies greatly, many forms having colored foliage, and variously shaped leaves. The colors are mostly tones of red and whitish green, the hues ranging from dark maroon to bright pink while there are some with bronze and golden tones. Usually the coloring appears as irregular strips along the line of the veins. Such red foliage plants usually have cerise flowers and red berries. ([Plate XVI](#))

POTHOS. TARO VINE. PHILODENDRON ***Scindapsus aureus* Linden**

(*Pothos aureus* Linden)

The huge green and gold leaved creepers which envelop so many coconut and other trees are called Taro Vine or Pothos by the local people while they are often called Philodendron by newcomers. The real Philodendron and Pothos are closely related and once were botanically mixed. Probably the name Taro Vine was applied because the large leaves suggest those of the Taro, which is also a relative. Pothos clings to its support with strong, woody roots which, however, do not draw nourishment from the host plant, since it is not a parasite. In the sun, under normal conditions the leaves are marked with gold, but when the vine grows in the dark they remain green and small, seeming almost a different plant. In this form it is often grown in water indoors.

It is a member of the Arum family and a native of the Solomon islands. ([Plate XVI](#))

CALADIUM. VARIEGATED-LEAVED A'PES

***Caladium bicolor* Ventonat**

Shady tropical gardens often make use of the colored Caladiums to give color and they are also frequently seen growing in pots. The heart-shaped leaves of these plants are marked with red and light green in almost endless variations and designs, so that they become a specialist's hobby. Some are blotched with red and white; others have designs that are as fine as lace. The plants grow from tubers, several leaves pushing up on long slender stems. During part of the year these leaves die back and the plant rests. Caladiums are members of the Arum family and natives of tropical South America. ([Plate XVI](#))

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MONSTERA VINE

***Monstera deliciosa* Liebmann**

Large curious leaves with many natural holes in them, characterize the Monstera vine which is often seen in Honolulu. On mature plants the leaves are very large, thick, green and glossy. They tend to a pinnate form, and holes in the leaves continue the openings between the veins. The plant grows slowly, clinging by aerial roots to a support. Some of these roots hang down string-like, toward the ground.

Under favorable conditions the plant bears large, pinkish flowers, in general form like those of the A'pe illustrated on [Plate XVI](#). The spathe is large, pinkish, thick, soft and flesh-like in texture. It covers the spadix like a tent, but falls off in a few days. The spadix holds the inconspicuous, densely packed flowers which develop into many small berries, closely packed together, so that the entire fruit, when mature, resembles a large greenish cone. Its flavor is thought to be like that of pineapples and bananas, probably inspiring the specific name of *deliciosa*.

This plant is a native of Mexico and Guatemala and a member of the Arum family. ([Plate XVI](#))

RHOEO. TRADESCANTIA

Rhoeo discolor Hance

Stiff rosettes about a foot and a half high, made up of pointed leaves which are purplish-red below and green above are the Rhoeo or 102 Tradescantia. This little plant grows so easily it is seen in many gardens, adding color to shady corners. In the axils of the leaves appears a boat-shaped, spathe-like growth in which are the small white flowers. Each has three sepals and three petals. The Rhoeo is a native of Mexico and the West Indies. It is a relative of the Wandering Jew and belongs to the Commelina family. ([Plate XVI](#))

A'PE

Alocasia macrorrhiza Schott

Very large, heart-shaped leaves, some of them on stems four or five feet high, grow up from the rootstalk of the A'pe plant. The name (pronounced Ah-pay) is a Hawaiian word first applied to a native species which has glossy, green leaves and greenish yellow flowers. Another A'pe (the one illustrated on [Plate XVI](#)) has dull leaves and pinkish flowers. The strange flowers are a foot long and have an unpleasant odor.

The A'pes are closely related to the Taros (Colocasias) which are the principle food plant of the Hawaiian people. The thick root stalk of the taro is boiled and mashed, to become *poi*.

Footnotes

[1]

This is the name by which the tree has been known in Hawaii for many years,
through erroneous determination. *C. nodosa* is a very different tree.

[2]

Costus spicatus is another plant but the name has sometimes been mistakenly applied in Hawaii to Crepe Ginger.

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